System Documentation.

A comprehensive description of the windows server system.

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1. Introduction

A system doccumentation is collection of comprehensive documents describing the inner workings of a system (Amerssonis, 2024).This document explains how the Windows Server has been configured. It is part of the practical assignment for the Windows Administration I (IT377G) course. The goal is to provide students (peers) with an understanding of how the windows server was configured in this course, and also provide a step to step guide to performing some of the components within the server.

The guide includes details on setting **come back and list all the how to guides created**, configuring networks, and applying security settings using Group Policy Objects (GPOs). It also explains how to manage users and groups to ensure secure and organized access to resources. How-to guides provide simple, step-by-step instructions for tasks such as configuring DHCP, setting up GPOs, and managing remote desktop access.

A key focus is on understanding the role of Domain Controllers (DCs) and Read-Only Domain Controllers (RODCs), ensuring they are correctly placed and configured for security and efficiency.

By following this documentation, peers will learn how to create a stable and secure server environment, gaining practical skills for future IT tasks.

2. system overview

2.1. list of servers

Server name	IP address	Instaled services/role
site1-server	192.168.1.100	Active Directory, DNS,DHCP
site1-filesvr	192.168.1.200	Windows deployment service, windows server backup, file server for shared resources
site2-server	192.168.3.100	Active Directory (Replica , DHCP, DNS
Site2- backup	192.168.3.200	Backup folders
MGM01	192.168.3.60	Windows Administration Center

2.2. Networking description

2.2.1. IP networks for each site

Site	IP address
Site 1 servers	192.168.1.0/24
Site 2 servers	192.168.3.0/24
Site 1 clients	192.168.0.0/24
Site 2 clients	192.168.2.0/24

Run ipconfig /all on command prompt to see the network description for the computer. For example, figure 1 and figure 2 below shows the network description for site1-clients and site-2 clients respectively.

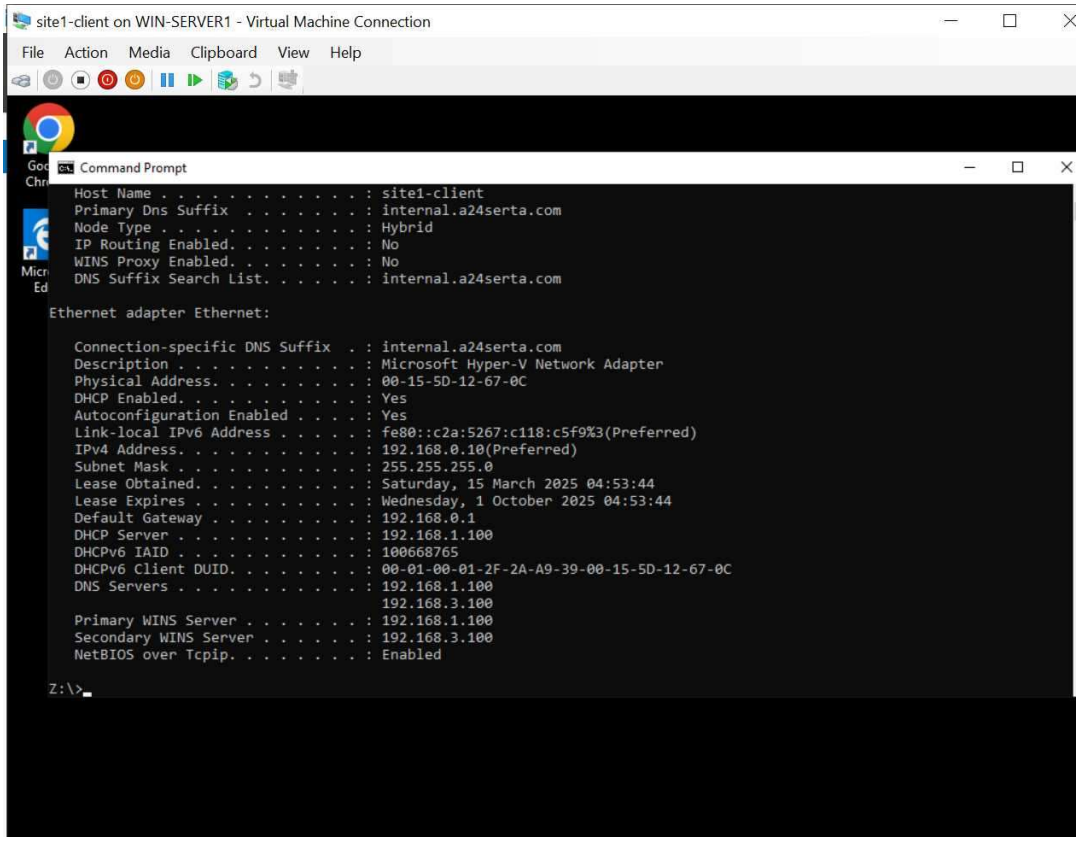


Figure1: network description for user (joka) on site1-client

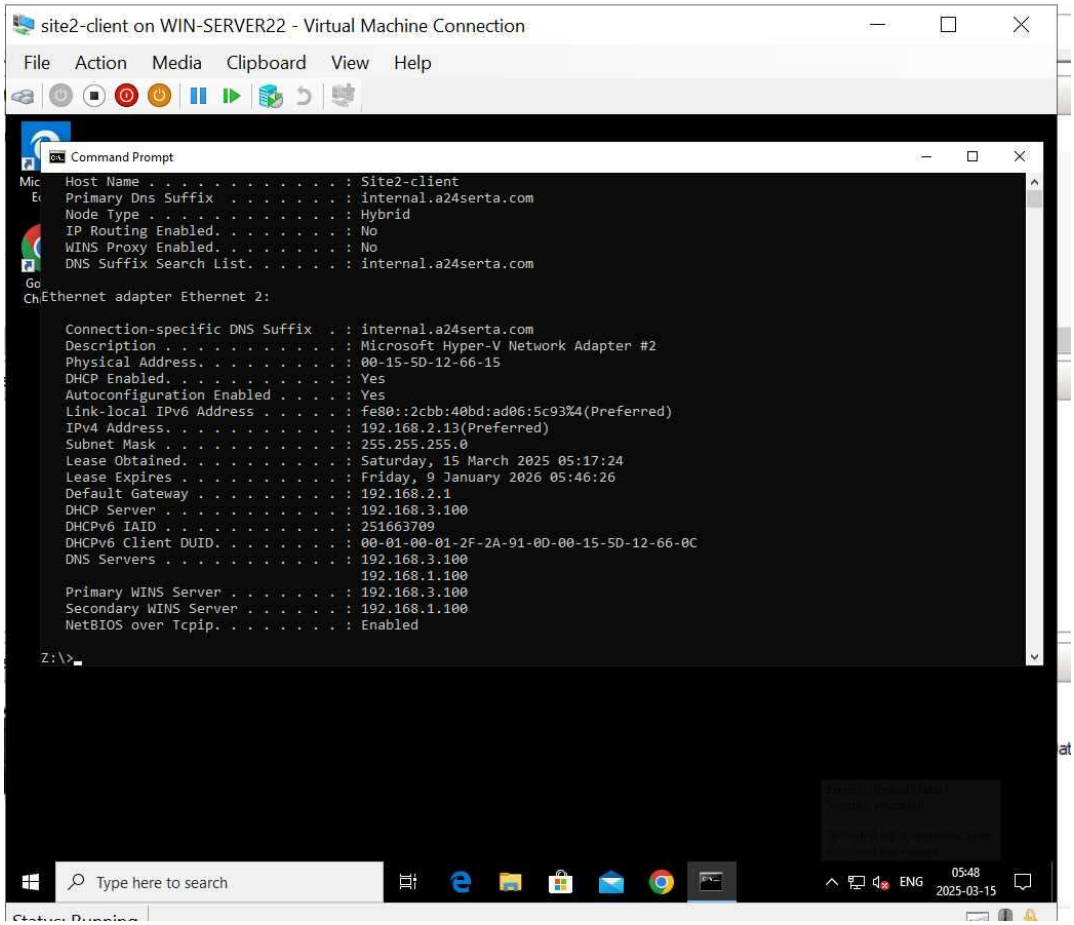


figure2: network description for user (sopa) on site2-client

2.2.2. DHCP scope and Scope options

Scope option	Site 1	Site 2
range	192.168.0.10-192.168.0.100	192.168.2.10- 192.168.2.100
Default gateway	192.168.0.1	192.168.2.1
DNS and WINS server	192.168.1.100	193.168.3.100

Figure 3 & 4 below are the snapshots of the dhcp scope options configured on both servers

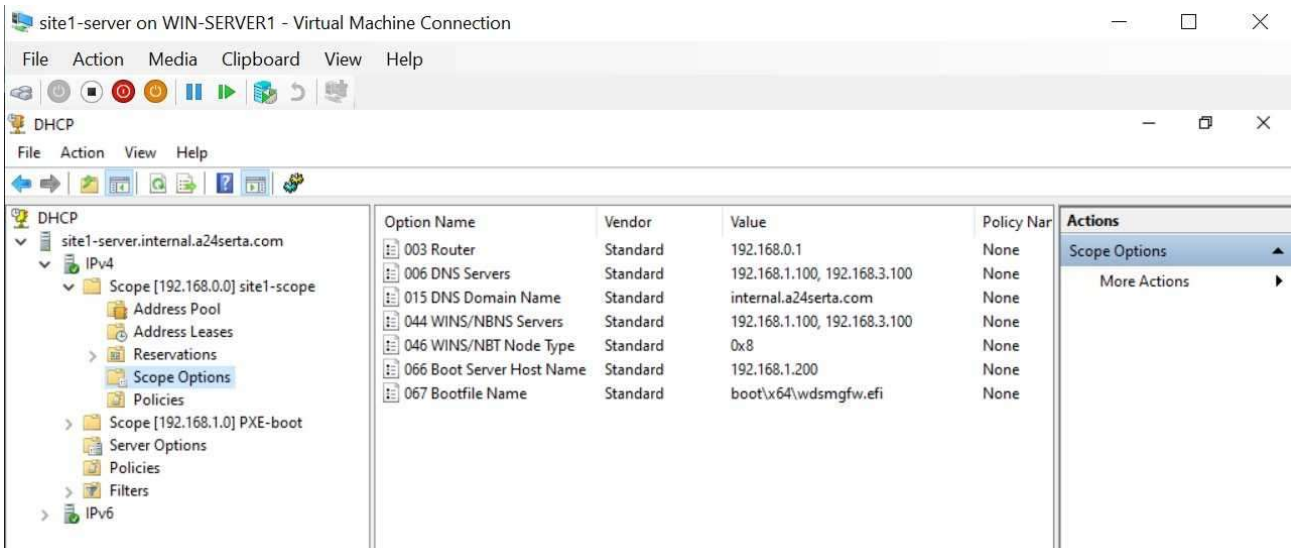


figure3: site1-server DHCP Scope Options

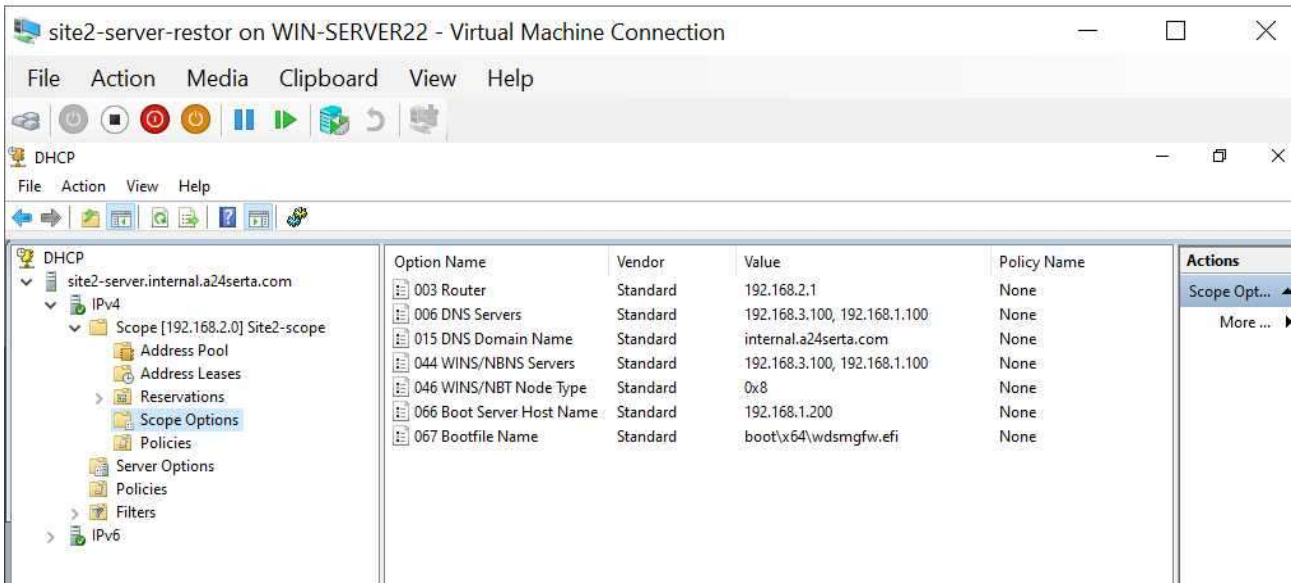


figure4: site2-server DHCP Scope Options.

2.3. description of design choices

2.3.1. Naming policies.

A consistent naming policy was used for the computers, the role groups and rule groups as follows ;

A) **Users:**

The users were imported from the users.csv files. Each user in the file has 2 names (which we can consider as the firstname and surname). The user name in this file is derived from the first 2 letters of both names. For example, Sofia Palmer's username is sopa.

B) **Organizational units:**

Birath & Ingemarsson, (2024) define an organizational unit as "*A logical container object used to organize other AD objects according to business needs, to add security or configuration control (using GPOs), or to delegate administrative control over the objects within an OU*". The OUs created in the domain ***internal.a24serta.com*** are ***My_Users, Role_groups and Rule_groups***. These descriptive names are based on the contents of the OUs. My_Users OU contains all the users imported from the user.csv file. The Role_groups contains the different departments for which the users have been classified under based on their responsibilities in the organization. These departments are *IT, Finance, RND, Executive and Sales department* (from the user.csv file). The term role simply refers to the function/responsibilities of the users in the organization.This is the last but not least OU created in my domain is the Rule_group. According to Microsoft best practice, permissions should never be assigned to individual users, instead security groups should be created and the users should be added to that group. Permissions can now be assigned to the group containing the users. The Rule_groups ou. contains the Role_groups to which permissions for resourses or access are ssigned to. The naming convention used for the rule_groups are;

- *For shared resources:* PermissionType-PermissionObject-permission,
- *For GPOs*. Rule groups that were used in the GPO settings, were given the same name as the gpos.

C) **GPO:** Microsoft, (n,d) defines group policy object as "a virtual collection of policy settings, security permissions, and scope of management (SOM) that you can apply to users

and computers in Active Directory." The gpos created have unique names as recommended by microsoft. The gpos were named based on the task they perform.

2.3.2. Group structure and design :

The groups are created using Roles and Rules. As earlier mentioned, the role_groups are created based on the departments and users are assignemt to them based on their roles/responsibilities. Rule groups are created based on specific permissions or access requirements. In sumary groups are created following the role based access control model (RBAC).

2.3.3. PLACEMENT OF GPOS.

Policy name	Placement
Allow_DRP_Access	My_Users
Disable_wallpaper_change	My_Users
Enable_firewall_Computers	Default Domain policy
RemoteInstall_7zip	My_Users
RemoteInstall_google chrome	My_Users
RemoteInstall_vlc	My_Users
Remove_bin	My_Users
set_wallpaper_calculator	My_Users
Show_hidden_files	My_Users
user_mapped_drives	My_Users

2.3.4. OU design.

In the domain, there are different ous for users, role groups and rule groups. Run the command dsquery ou in the command prompt to on any of the servers, to see all the OUs in the doma Figure 5 below is a snapshot of the different OUs in the domain.

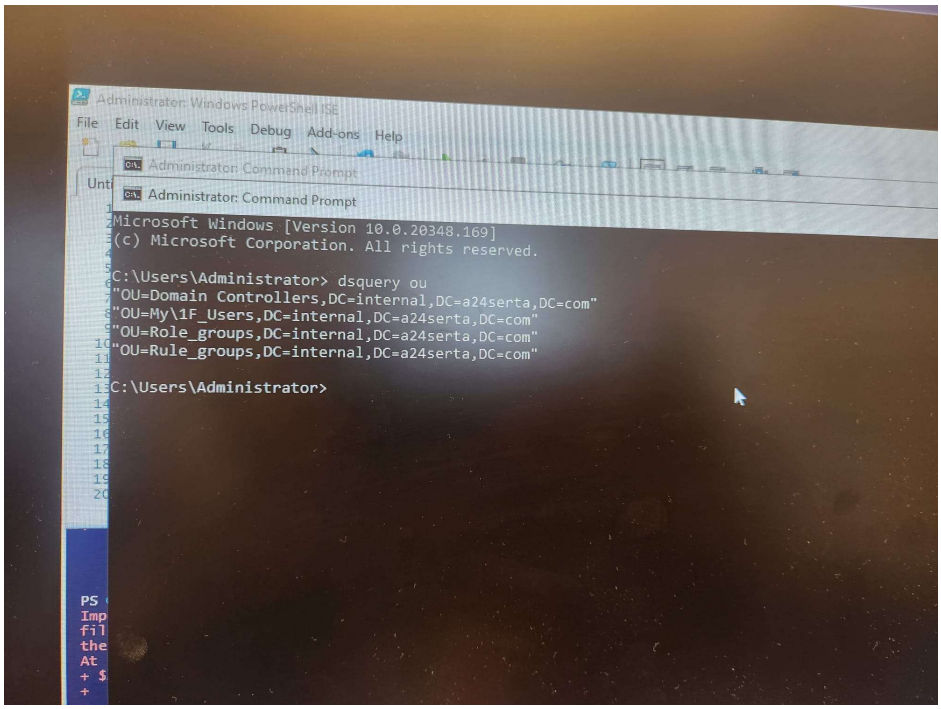


Figure5: screenshot showing the OUs created in the domain.

2.4. **Inventory of settings**

2.4.1. Existing gpos: All policies can be found under the group policy objects as shown in figure 6 below.

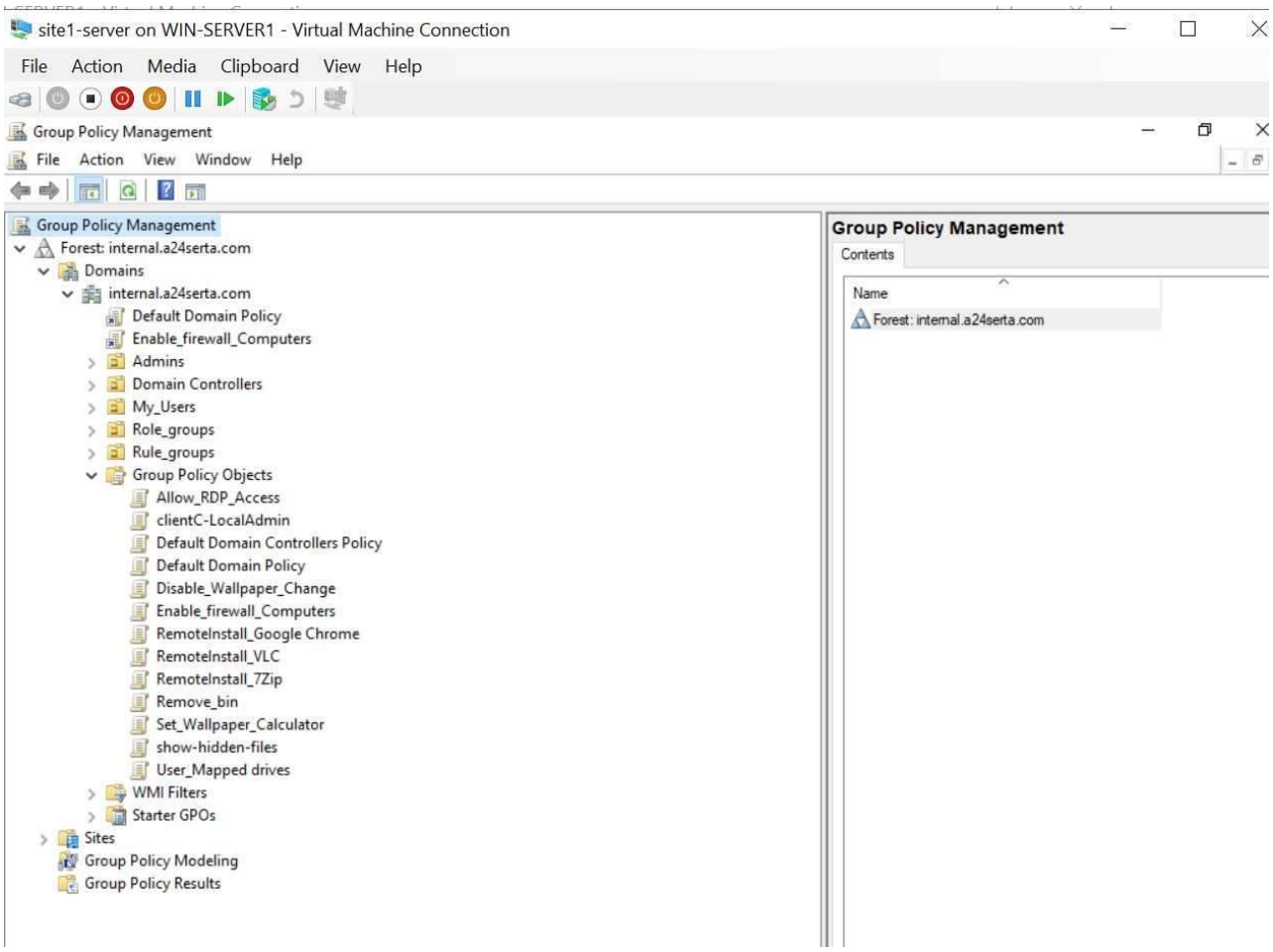


Figure6: Snapshot of all existing GPOs.

2.4.2. Delegations.

Members of the IT department have Remote Desktop access to the site1-server and site1-server.

A) Role_groups**:** all existing role groups are shown in figure 7.

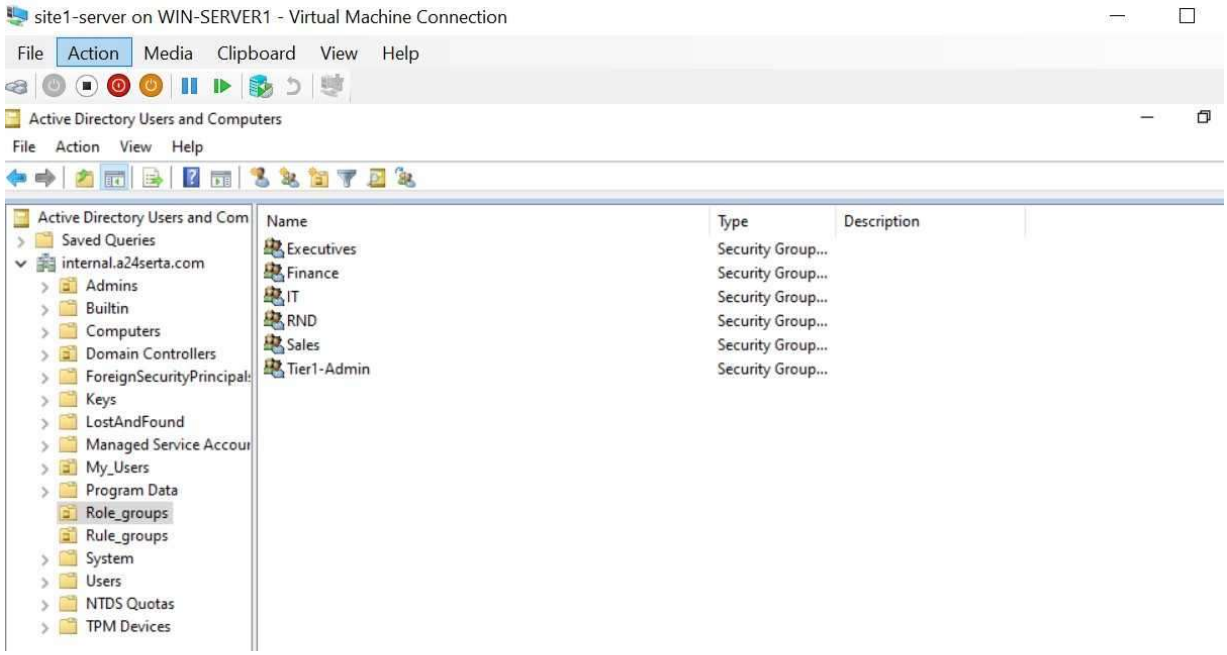


figure7: snapshot of the Role_groups.

B). Rule_group.

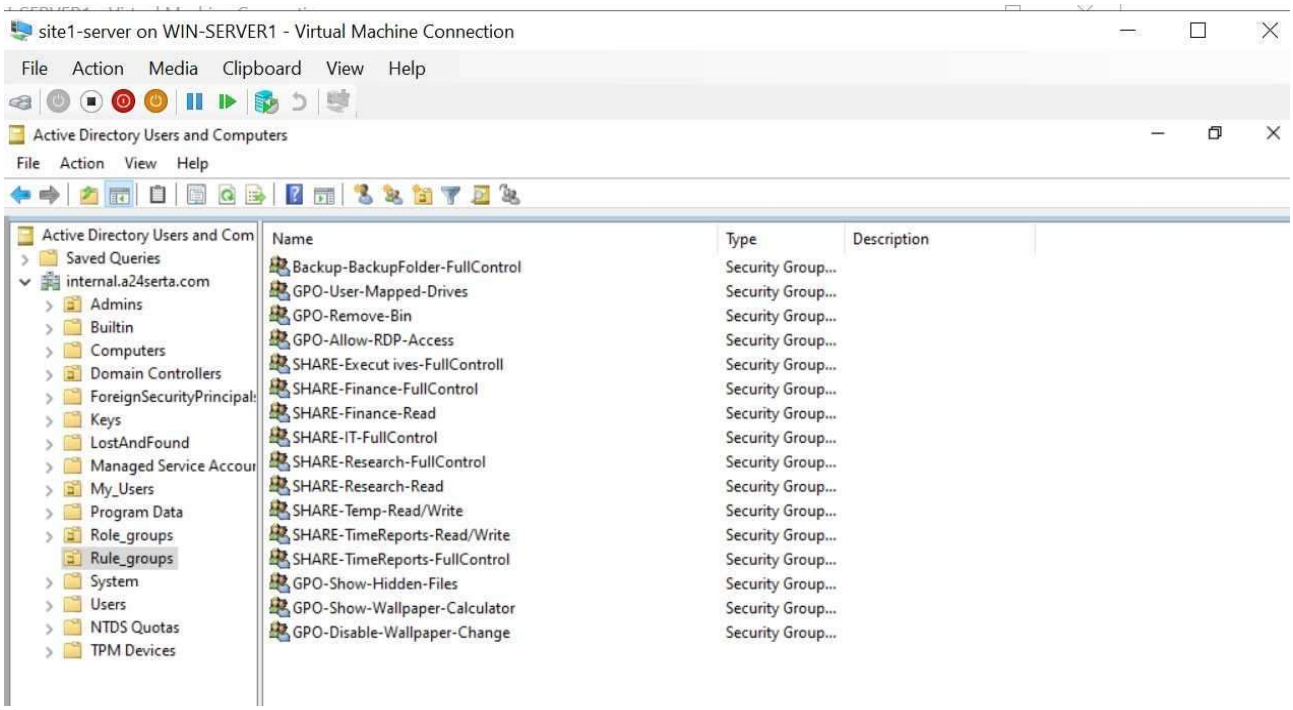


Figure8: Rule_groups

Name	members
GPO.Allow-RDP-Access	IT
GPO-Disable-Wallpaper-Change	All the Role_groups
GPO-Show-Wallpaper-Calculator	Finance
GPO-User-Mapped-Drives	All the Role_groups
SHARE-Finance-FullControl	Finance , Sales
SHARE-Finance-Read	RND, IT, Sales
SHARE-IT-FullControl	IT
SHARE-Research-Fullcontrol	RND
SHARE-Research-Read	Executives
SHARE-Temp-Read/write	All the Role_groups
SHARE-TimeReports-Fullcontrol	Executives
SHARE-TimeReports-Read/write	IT, RND, Sales, Finance.

3.Answers to demo questions

This practical assignment was divided into 10 steps including questions for each step. In this chapter, all the questions for each step will be answered accordingly. For the purpose of this exercise, the steps will be called section. Therefore, there are 10 sections , with answers provided for the questions under these sections.

Section 1:Building virtual machine

1. What is the difference between Internal and External switches in Hyper-V?

To bigin, it is important to understand what a hyper-V virtual switch is. Hyper-V Virtual Switch is a software-based layer-2 Ethernet network switch that is available in Hyper-V Manager when you install the Hyper-V server role (microsoft, nd). According to (Zacker et al., 2012), there are 3 types of virtual switches that is internal, external and private switches.

External Switch: As explained by (Zacker et al., 2012), this switch is Connected to the physical network via the host's network adapter. It allows both the host (parent partition) and virtual machines (child partitions) to access the external network.

Internal Switch: the internal switch is not connected to the physical network, but the host can still access the physical network (Zacker et al., 2012). this switch enables communication between virtual machines and the host OS, but child partitions cannot access the external network (Zacker et al., 2012).

Private Switch: According to Zacker et al., (2012) the private switch is isolated within the Hyper-V server and it only allows communication between virtual machines on child partitions. The host OS cannot access this network (Zacker et al., 2012).

The table below state the difference between the internal and external switch with regards to how the network is build, the physical network access and the scope of connectivity.

FEATURES	INTERNAL VIRTUAL SWITCH	EXTERNAL VIRTUAL SWITCH
Network building	Bound to a separate instance of the networking protocol stack in the host OS, independent of the physical network interface adapter.	Bound to the networking protocol stack in the host OS and connected to a physical network interface adapter.
Physical network access	Virtual machines in child partitions cannot access the physical network; only the host OS in the parent partition can.	Virtual machines in both parent and child partitions can access the physical network.
Connectivity scope	Only provides connectivity between virtual machines and	Provides connectivity to both virtual machines and external

	the host OS and not to external networks.	networks.

***Source**: created based on the information provided by Zacker et al., (2012), lesson 9.*

2. What is the difference between the Datacenter and Standard editions of Windows Server and when is each type preferable?

FEATURES	WINDOWS SERVER DATACENTER EDITION	WINDOWS SERVER STANDARD EDITION
Intended use	Designed for large, highly virtualized environments.	Suitable for physical environments with limited virtualization.
Virtualization support	Allows unlimited virtual machines (VMs).	Allows only two virtual machines (VMs).
cost	More expensive due to its advanced features.	More affordable with standard features
Key features	Includes additional functionalities like storage spaces direct	Includes traditional server features with fewer advanced capabilities.
When to use preferably.	Ideal for organizations requiring extensive virtualization.	Best for businesses needing minimal virtualization or mostly physical setups.

Source: the table was created based on Birath & Ingemarsson, (2024).

Section 2: create a new domain for site 1.

1. Why do you need a DNS-server in an AD domain?

DNS plays a very important role in Active Directory (AD) environments, providing essential services for network functionality and Active Directory communication (Zacker et al., 2012). Therefore it is essential to have a DNS server in an Active Directory (AD) domain. Here's an overview the key functionsof the DNS server

Name Resolution: It translates computer names or Fully Qualified Domain Names (FQDNs) into IP addresses, allowing domain-joined computers to locate domain controllers and other network resources (Zacker et al., 2012).

- **Reverse Lookup:** It enables reverse lookups, converting IP addresses back into computer names, which is useful for network management and troubleshooting (Zacker et al., 2012) .

- **Service Location:** AD relies on DNS to locate and communicate with domain controllers, global catalog servers, and other AD services using special DNS records (e.g...SRV, A, AAA), (Zacker et al., 2012).
- **friendly user- access**:,DNS allows users to connect to computers and services using readable names instead of complex IP addresses, thereby improving ease of use and efficiency in an AD domain (Zacker et al., 2012).

2. What forwarder is your DNS-server configured to use?

The DNS server is configured to use 10.0.252.201 as its forwading adress as shown in figure 9.

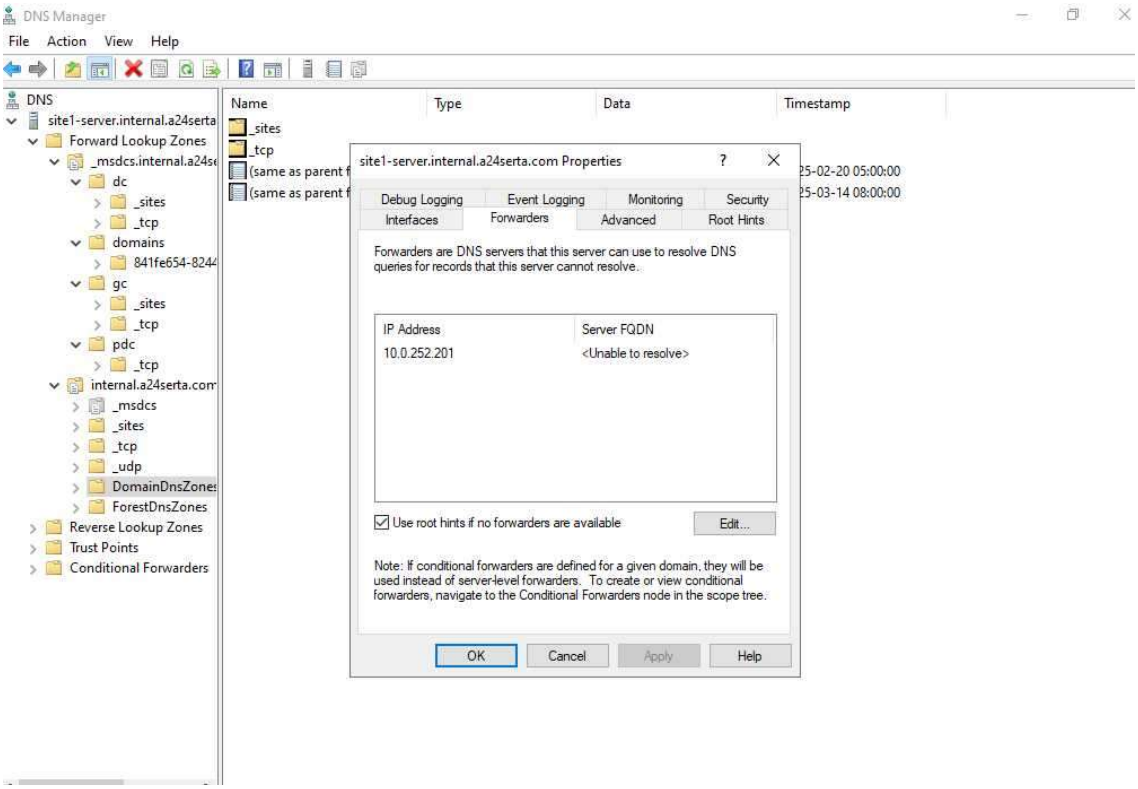


figure9: snapshot of the DNS-server fowarding adress.

3. What effect does the forest functional level have?.

The forest functional level (FFL) in Active Directory determines the set of advanced features available for the entire forest and the minimum Windows Server versions that can be used for domain controllers within that forest (Microsoft. n.d). For this lab exercise, the servers forest functional level was set to *2016* which implies that only domain controllers running *Windows Server 2016 or higher* can be added to the forest. Also, advanced features specific to Windows Server 2016 are available, enhancing security and directory services and all the domains within this forest must also have a domain functional level of at least 2016. it is important to note that the FFL set doesn't affect the operating systems of workstations or member servers, (Microsoft. n.d).

Section 3: structure and populate your AD.

1. What is the purpose of using nested groups?

Nested groups in Active Directory are structured approaches involving putting a group in another group (Birath & Ingemarsson, 2024). Microsoft best practice recommends using nested groups in AD as it provides several important purposes as stated below.

- **For easier permission management**: Using nested groups simplifies assigning permissions this means that if administrators grant access to a primary group, and all subgroups and members will automatically inherit those permissions (Mutama, 2023). This approach reduces the need to manage each user or group separately.
- **Structured Organization**: Mutama (2023) also states that nested groups help build logical hierarchies that align with an organization's structure, makes it easier to manage and control access based on roles, departments, or management levels.
- **Less Administrative Effort**: Managing user access becomes more efficient with nested groups therefore, instead of updating permissions for each individual, changes can be made at the group level, saving time and effort (Mutama, 2023).
- **Better Access Control**: according to Mutama, (2023), nested groups provide more precise control over who has access to resources, making it easier to follow the principle of least privilege.
- **Simplified Auditing**: With permissions centralized through nested groups, auditing user access becomes simpler and more straightforward (Mutama, 2023).
- **Efficient Group Management**: Last but not least, Mutama (2023) assert that nesting reduces the number of individual groups with similar permissions, keeping Active Directory organized and easier to manage.

2. Why is it a bad idea to give permissions to individual users?

According to best practices, users should be added to a group with role-based access control. The users then inherit the permissions of the group they are in (Birath & Ingemarsson, 2024). It is therefore a bad idea to give permissions to individual users because of the several reasons. According to Birath & Ingemarsson (2024), when permissions are given directly to each user, it becomes challenging to keep track of who has access to what. This can quickly turn into a complex and confusing process as more users are added. VC3 (n.d.), assert that when users are granted individual permissions, there's an increased chance of accidental data exposure. Users with excessive or unnecessary access may unintentionally compromise sensitive information due to limited cybersecurity awareness or mistakes (VC3. n.d.). Another reason why it is a bad practice to give permissions to individual users is the fact that directly assigning permissions increases the

chance of human error (Birath & Ingemarsson (2024). For example, it's possible for administrators to forget revoking access when someone changes roles or leaves the company, potentially leading to unauthorized access. As organizations grow, managing individual permissions becomes more difficult, (Birath & Ingemarsson 2024). VC3 (n.d.) explain that establishing a group-based or role-based access control model from the start ensures a more sustainable and scalable approach. Birath & Ingemarsson, (2024) also state that managing permissions through groups allows for easier updates as adding or removing someone from a group automatically adjusts their access, reducing the need for manual changes and helping prevent security risks.

3. Why should administrators use a regular user account for everyday tasks?

- For accountability and auditing. It is easier to know who made changes to the system when a regular user account is used for everyday tasks, when reviewing the system logs, as compared to when the default administrator account is used (Birath & Ingemarsson 2024).
- To reduce security risk. A regular user account follows the least privilege principle which implies the impact of accidental actions, such as deleting critical files or changing system configurations can greatly be minimised (Birath & Ingemarsson 2024).

4. What is the difference between AGDLP and AGUDLP and when would each be applicable?

AGDLP and AGUDLP are both group nesting strategies (Birath & Ingemarsson, 2024). AGDLP (Accounts → Global → Domain Local → Permissions) is the acronym for the order in which nested group strategy should be implemented. The U in AGUDLP stands for universal, which determines the difference between the 2 strategies. AGDLP contains users, computers local and global groups from the same domain (Birath & Ingemarsson, 2024). Birath & Ingemarsson, (2024) also highlighted that AGDLP is used to manage resources in any domain under the same forest and permissions are only granted to groups within the same domain. According to the authors, AGDLP is best applicable for a single-domain environment. However, AGUDLP contains users, computers global groups and universal groups from any domain within the same forest (Birath & Ingemarsson 2024). It can also be used to assign permissions to multiple domain local groups provided they are in the same forest. AGUDLP is best applicable in a multi-domain environment in an AD forest.

Section 4: configure fileserver

1. How does share permissions and NTFS permissions work in relation to each other?

According to Schrader, (2025) in his blog "t*he complete guide to NTFS vs share permissions*" share permissions and NTFS permissions work together to control access to shared resources, but the more restrictive permission takes precedence. Share permissions apply when accessing the folder over the network, while NTFS permissions apply locally on the system (Schrader 2025; Birath & Ingemarsson 2024). If both permissions are set, the final access level is determined by the most restrictive permission between the two. For example, if a folder has Full Control share permissions but only Read NTFS permissions, the user will only have Read access because NTFS permissions are more restrictive. It's best to set share permissions at a higher level for general access and use NTFS permissions to fine-tune access for specific users or files within the shared folder. This way, users can easily work with most of the files, but sensitive data can be protected by stricter NTFS controls.

2. Explain the purpose of the CREATOR OWNER placeholder.

The CREATOR OWNER placeholder in Windows Server is used in NTFS permissions to automatically assign ownership and specific rights to the user who creates a file or folder (Birath & Ingemarsson, 2024). This implies that the person (user) who creates an object (folder, subfolders or files) is granted certain permissions over it, regardless of broader permissions set for other users or groups (Birath & Ingemarsson, 2024). The creator owner placeholder is an efficient way to automatically assign administrative rights to object creators, ensuring they can manage their own resources without manual permission adjustments (Birath & Ingemarsson, 2024).

3. Describe how permission inheritance on folders works

In Windows Server 2022, permission inheritance simplifies access control by automatically applying the permissions set on a parent folder to its subfolders and files Microsoft. (n.d). This default action ensures consistency across the directory structure and makes it easier to manage permissions (Birath & Ingemarsson 2024). Users can view these inherited permissions by accessing the folder's security settings, where the "Inherited from" column indicates the source of the permissions. If necessary, inheritance can be disabled for specific folders or files. When doing so, there are two options to choose from either converting inherited permissions into explicit ones, which keeps the current permissions static even if changes occur at the parent level, or removing inherited permissions entirely to allow for custom access settings (Birath & Ingemarsson, 2024). Once inheritance is disabled, administrators can manually assign permissions to meet specific access requirements.

4. Describe how you create home folders for multiple users at once in Active Directory.

Before creating home folders for the users, ensure that the home folder is created and accesible to all the users over the network by sharing the folder. For this exercise, the HomeFolders are created in the site1-filesvr and the share path is \\SITE1-FILESVR\HomeFolders. Follow the steps to create homeFolders for all the users simultaneously in AD.

*step1:*To create home folders for multiple uses, Open active directory Users and computers and navigate to the OU containing the users (My_Users OU) then double click to open.

Step2: select all the users , right click and open properties. This step ensures any change made in the properties settings is applied to all the users.

Step3: Inside the properties settings, click on the profile tab, click connect under the homefolders section and assign a unique drive letter.

*step4***:** specify the path of the homefolders (\\SITE1-FILESVR\HomeFolders\%username%). The %username% variable is used so that unique folders can be created for each user. Click okay to apply.

Section 5: Configure clients using GPO.

1. Elaborate on why you want to be able to make settings from a central point.

Managing computer settings from one central point offers several advantages, especially for organizations with multiple offices or remote workers. According to a post by Rest solutions (n.d), tittled *´´Benefits of a centralized computer management´´*, managing computers from a central point simplifies tasks, improves security, and recduce cost. When settings are controlled centrally, it becomes much quicker and easier to manage computers. Rest solutions (n.d) state that updates, settings changes, and troubleshooting can all be done remotely, saving time and avoiding the need for on-site visits. Futhermore, This approach also ensures that all systems are set up consistently, reducing the chances of software or compatibility issues (Rest Solutions, n.d). In terms of security, centralized management makes it easier to apply updates and enforce safety protocols across all devices. In a centralised location, administrators are able to monitor systems for unusual activity and address potential threats before they escalate. Additionally, managing systems remotely can lower costs by reducing the need for travel, cutting hardware expenses, and minimizing maintenance costs (Rest Solutions, n.d). Overall, central management helps organizations maintain smooth, secure, and efficient operations, no matter where their employees are located(Rest Solutions, n.d) .

2. How can you, client side, check what GPOs that are applied to a user?

Clients can check the GPOS applied to them by running the command **gpresult /r** in command prompt

3. How can you, client side, force an update of GPOs?

- *Open command prompt and run **gpupdate /force***
- Restart the computer for the configurations to apply.

4. Describe the different ways to configure what users a group policy is applied to (i.e. linking, security filtering, and item-level targeting).

Group Policy Objects (GPOs) in Active Directory can be configured to apply to specific users or computers through various methods:

1. Linking

Linking is the basic method of applying a GPO to users or computers (Microsoft. n.d). For your GPOs to work properly, they must be linked under the right OU. This can be achieved by simply creating the GPO under the right OU in the Group Policy Management console. However, if the GPO is created in the group policy object container or user a wrong OU, we can link it to the desired OU;

- Right-click on the desired Organizational Unit (OU) or domain and click "Link an Existing GPO". Choose the GPO you want to link.
- The GPO will now apply to all users and computers within that OU or domain.

2. Security Filtering: Security filtering allows administrators to set the GPO application to specific users, computers, or groups this is done by following these steps;

NB. a test gpo was created to show the steps

- In the GPMC, sellect and click the GPO, an editor page will open.
- Find the "Security Filtering" section under the Scope tab in and click to specify the users, computers, or groups that should receive the GPO. See figure 10 below

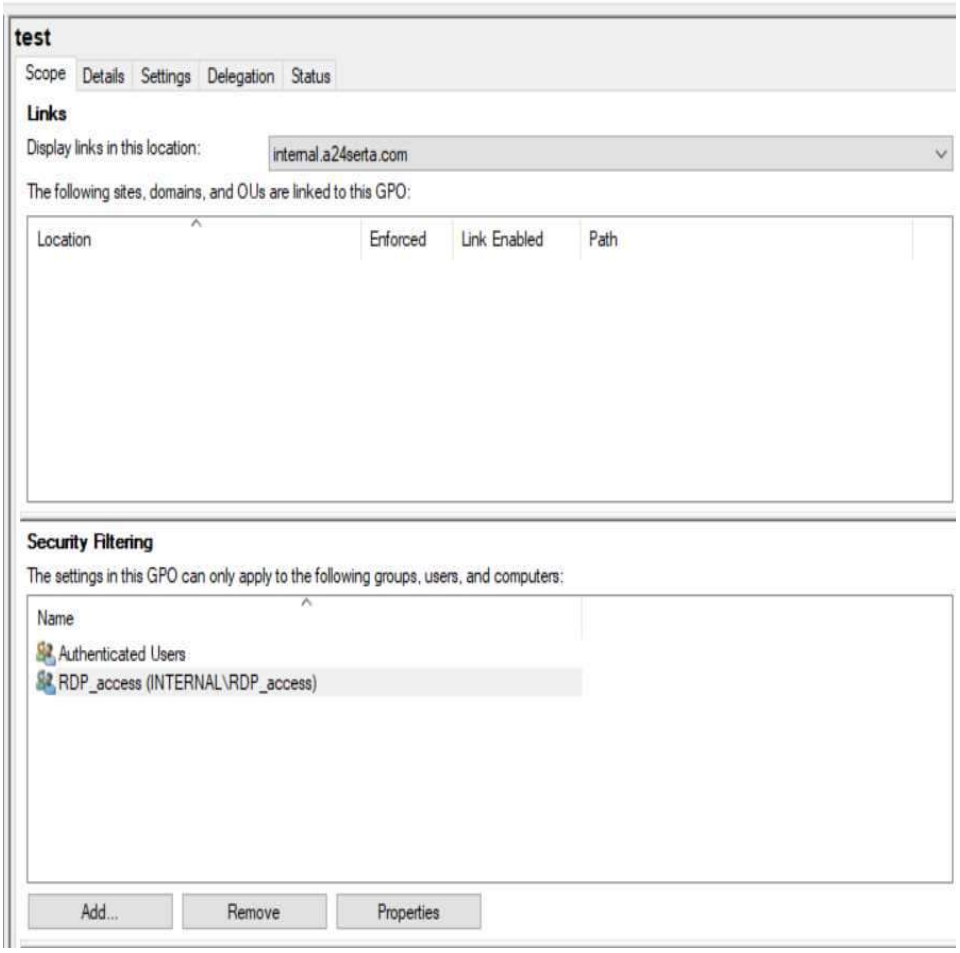


Figure10: snapshot showing security filterrng

- Even though Microsoft says to Remove "Authenticated Users" if you want to restrict the GPO, it is not a good practice to do so as authenticated users is a default domain group and removing it completely caun cause your gpo not to apply at all.
- After adding the target group, click on delegation and click advanced…, this will open the security settings for the gpo where you can manage the permissions. If you want to apply the gpo to a specific group, click on authenticated useres and uncheck allow for ”apply group policy”.This action ensures that authenticated users have only read permission for that gpo.
- To ensure that the group or computer you have added have the right permission, open the delegation tab and click advance. Select the group, user or computer you added and verify that they have read and the appy group policy permissions checked.
- Click okay to apply. See the policy security settings tab in figure 11.

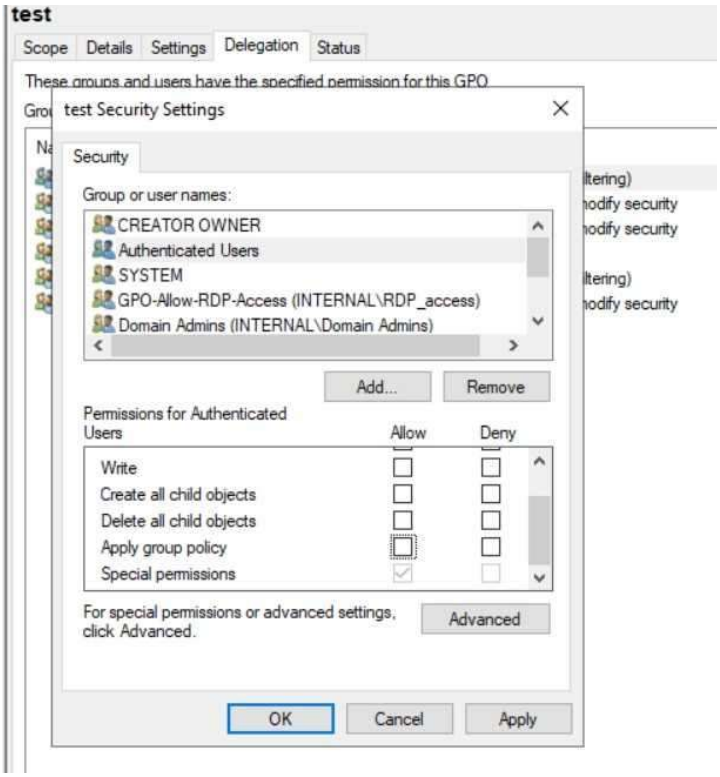


figure 11: security settings tab of the target policy (test)

NB:Once the appy froup policy permission has been unchecked for the authenticated users, that object is authomatically removed from the security filtering under scope as shown in figure 12. This method is useful for applying policies to specific groups without creating multiple Ous (Microsoft, n.d).

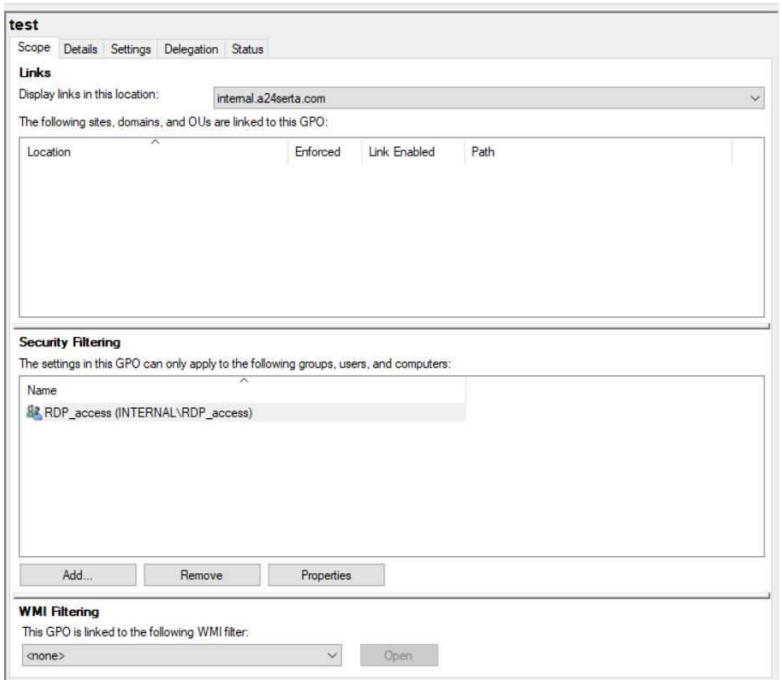


figure12: snapshot of security filtering after the allow apply group policy permission has been unchecked.

3. Item-Level Targeting (ILT): ILT provides granular control over individual preference items within a GPO (Microsoft, n,d).

- Right click on the specific GPO and select edit. Expand preference and navigate to the specific preference item and click okay within the GPO. This applies to both user and computer configurations. You can set item-level-targeting on a new or existing item. To create a new preference item, click the specific preference and click new. Give the name and set other parameters accordingly then click on the ”Common” tab
- for already existing preference items, right-click and open properties, select the "Common" tab. Check "Item-level targeting" and click "Targeting”

figure 13: snapshot of the ¨Common tab¨

- In the targeting editor, click on new item and choose the object. Type as shown in figure 14. Then add the object by clicking the (…). if you add morethan 1 object, ensure that item options is set correctly based on your requirements as shown in figure 15.

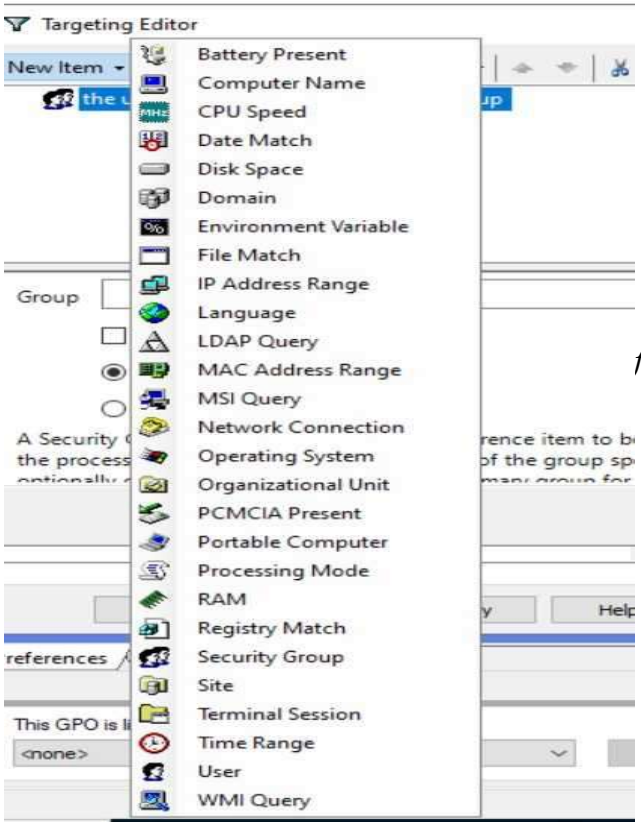


figure14: targeting editor

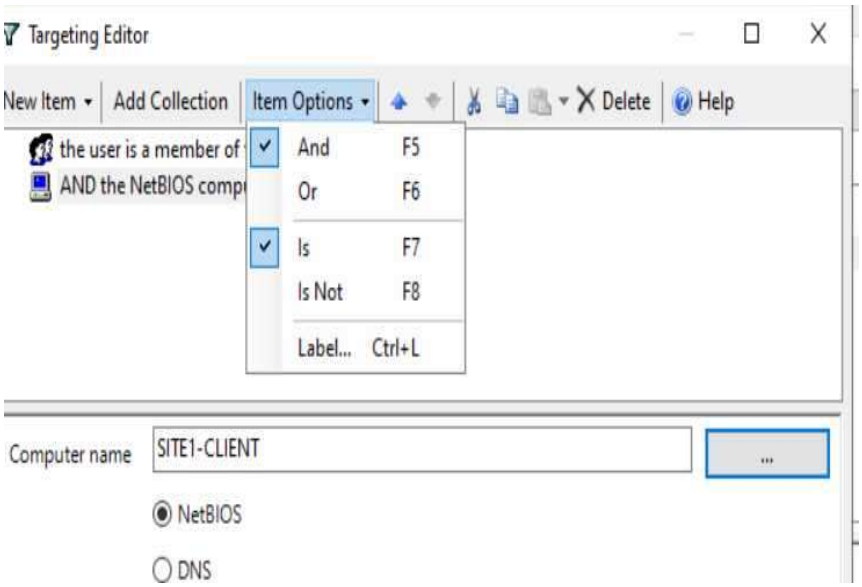


figure 15: targeting editor

Section 6: set up site 2

***1**. How can you, client side, check what logon-server you used?*

Open command prompt and run the command set logonserser. See the figure 16 and 17 below showing the logon-servers for site1 and site 2 clients respectively.

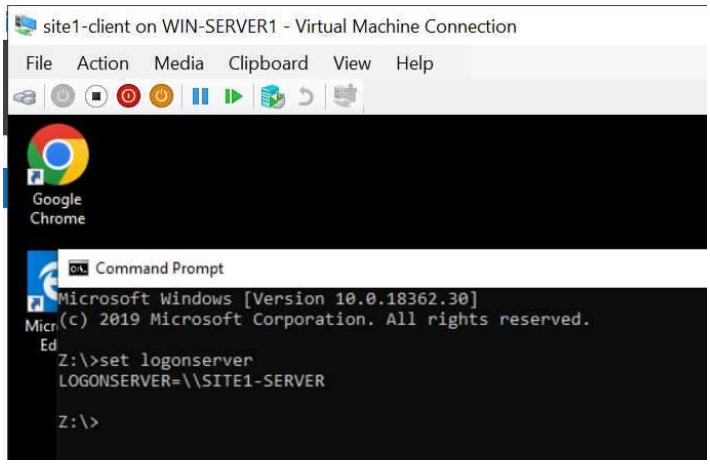


Figure 16: site1-client logon-server.

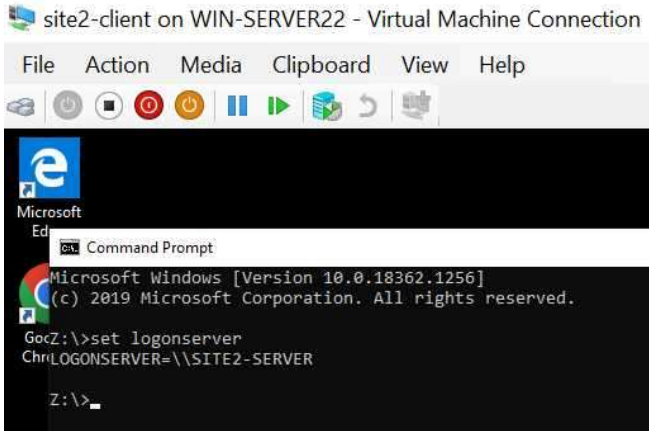


Figure17: site2-client logon-server.

2. Briefly describe how a Read-only DC works.

A Read-Only Domain Controller (RODC) is a type of domain controller in Windows Server that hosts a read-only copy of the Active Directory database (Microsoft, n.d). It contains all AD objects and attributes except account passwords (unless explicitly allowed). Also, changes cannot be made directly to the RODC; they must be made on a writable domain controller and then replicated back to the RODC (Microsoft, n.d). RODCs are ideal for remote or branch office environments with limited security, low bandwidth, and minimal IT expertise. They enhance security and provide efficient access to network resources while reducing the risk of unauthorized changes (Microsoft, n.d.).

3. Elaborate on the benefits of having multiple Dcs

Having multiple Domain Controllers (DCs) offers several significant benefits for network infrastructure. Firstly, it improves availability and redundancy, ensuring that authentication and directory services remain operational even if one server fails (Birath & Ingemarsson, 2024). Having mutiple DCs also improve load balancing, as authentication requests are distributed across multiple DCs, reducing strain on any single server and boosting performance. Additionally, placing DCs closer to users can speed up authentication processes, reducing latency and enhancing user experience. In terms of disaster recovery, multiple DCs facilitate quicker restoration of services and minimize data loss during failures. Security is also strengthened by deploying Read-Only Domain Controllers (RODCs) in less secure locations, as these controllers do not store password data by default. Furthermore, multiple DCs simplify maintenance and upgrades by allowing one DC to be serviced while others remain active, preventing service interruptions. Finally, having multiple DCs ensures faster replication and synchronization of directory data, maintaining consistency across the network, (Birath & Ingemarsson, 2024).

4. Are there any drawback of using multiple DCs that are all configurable

While multiple Domain Controllers (DCs) enhance redundancy and availability, there are some potential drawbacks when all DCs are fully configurable. One significant concern is the risk of data inconsistency. Since each writable DC can accept changes, incorrect configurations or errors made on one DC can be replicated across all others, potentially leading to widespread issues. Additionally, managing multiple writable DCs increases the administrative complexity, as careful coordination is required to ensure consistent configurations and avoid conflicts. Another challenge is the replication traffic generated between DCs, which can strain network resources, especially in environments with limited bandwidth. Security risks are also heightened, as compromising one writable DC can potentially allow an attacker to manipulate the entire Active Directory environment. Therefore, while multiple DCs provide benefits in terms of redundancy and load distribution, they also require careful planning and management to mitigate these potential drawbacks.

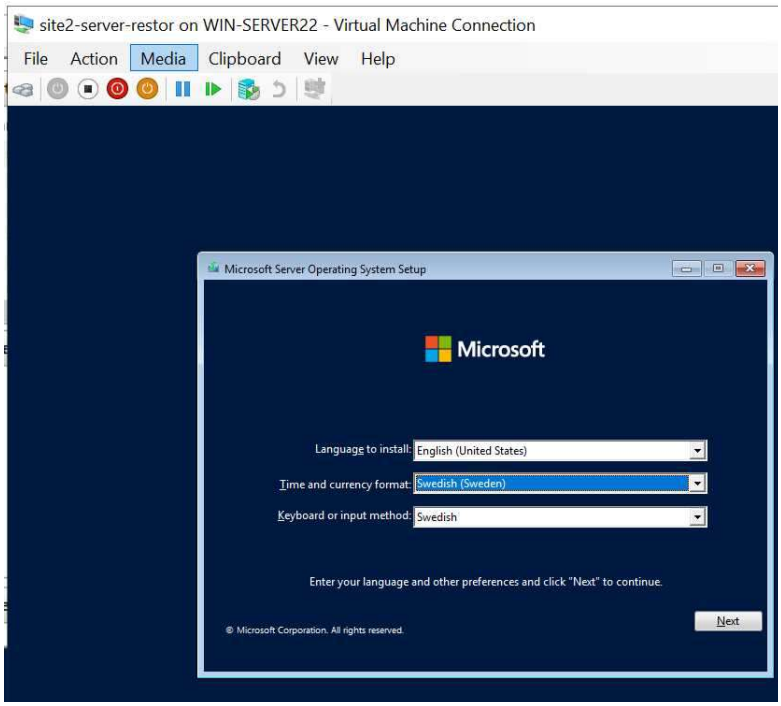
Section 7: configure backup

1. Create a how-to guide that describes in detail how to make a bare metal recovery of the Site-2 DC in your domain (pictures for each step is necessary).

This guide provides a step-by-step process for performing a bare metal recovery of the Site2-server. Before performing the bare metal recovery, some prerequisits have to be met; Ensure that you have a valid and recent backup of the entire server you wish to restor saved in a secured location. The backup was performed using the windows server backup tool. Also, your original server should be turned off.

To perform the metal recovery, begin by installing a new virtual machine with the same parameters like the original server. The server created is called ”site2-server-restor.

step 1:start the computer and press any key for to boot from the DVD Drive. When the media starts, set the language, time and key board same as your original server (swedish) and click next as shown in the figure 18 below.



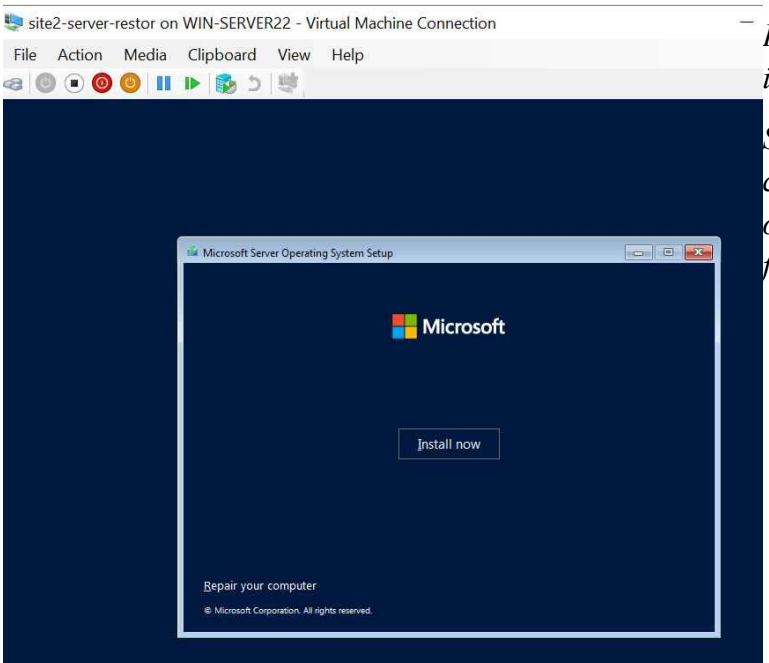
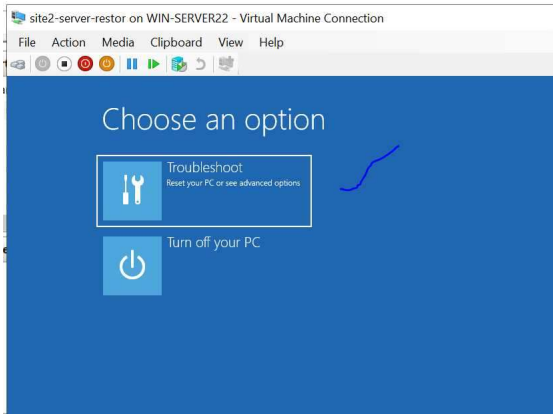


Figure 18: screenshot of the installation media.

Step2:next, click on repair your computer found on the bottom left of the screen as shown on the figure 19 below.

Figure 19: snapshot of the installation media.

Step3: next click on troubleshoot , then ckick on image system recovery as shown on fig 20 and 21 below.



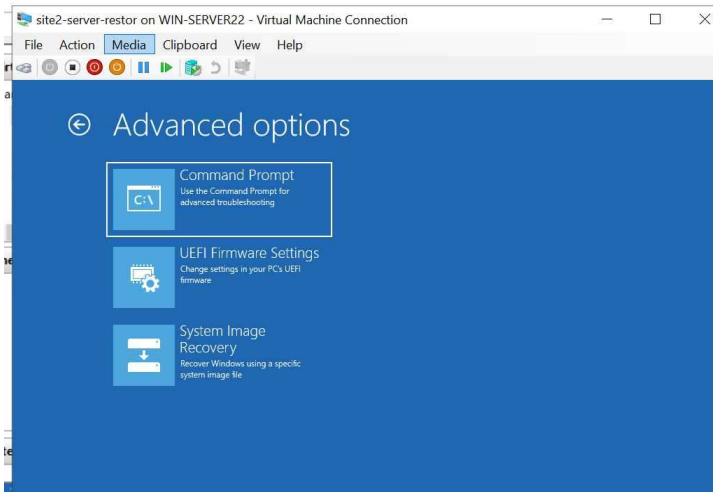


Figure 20: snapshot of insta media

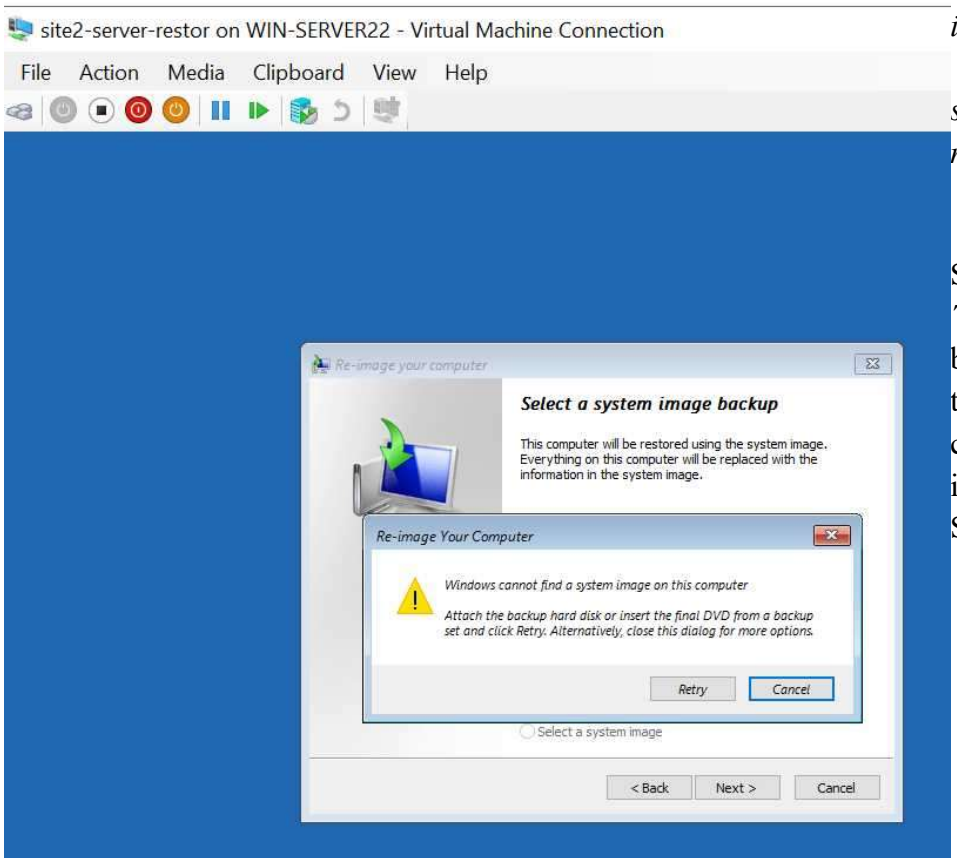


figure 21: snapshot of installation media

Step4. On the select a ´´system image backup" wizard, close the error message, click select a system image then click next. See figure 22 and 23

figure: 22. Prompt to select a system image backup.

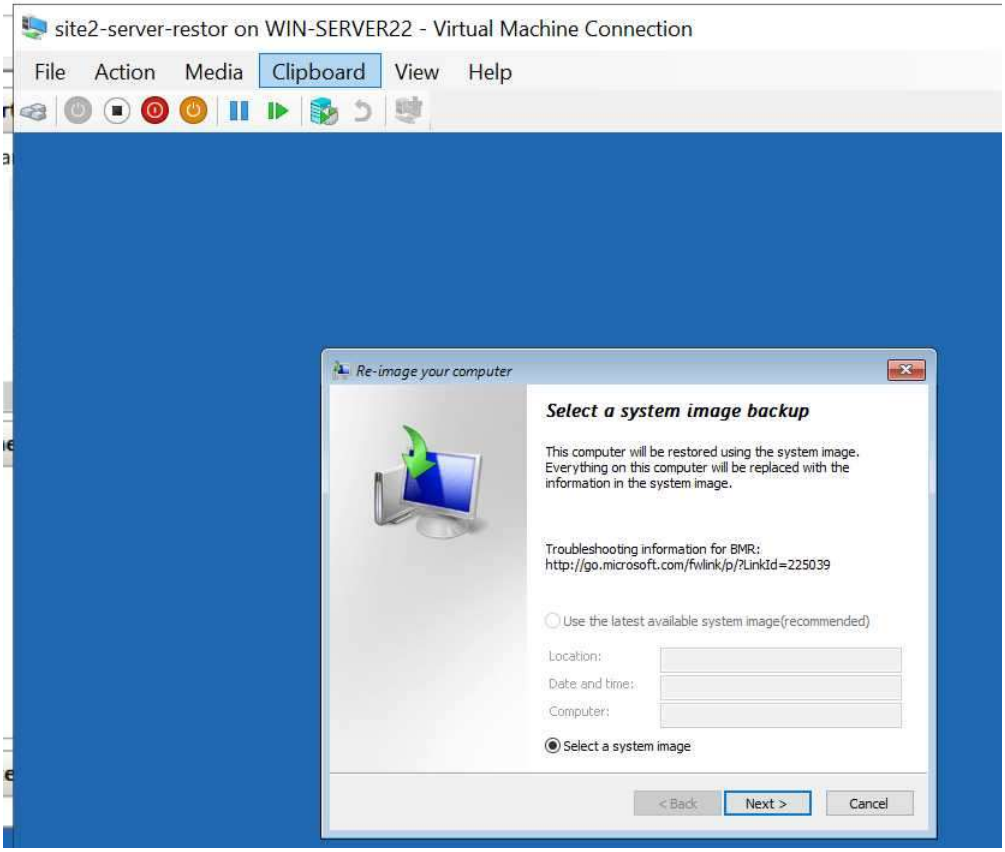


Figure 23.

step5: The wizard will then prompt to choose where to select the image, since the backup image is on a folder that has been shared over the network, click on search for a system image over the network

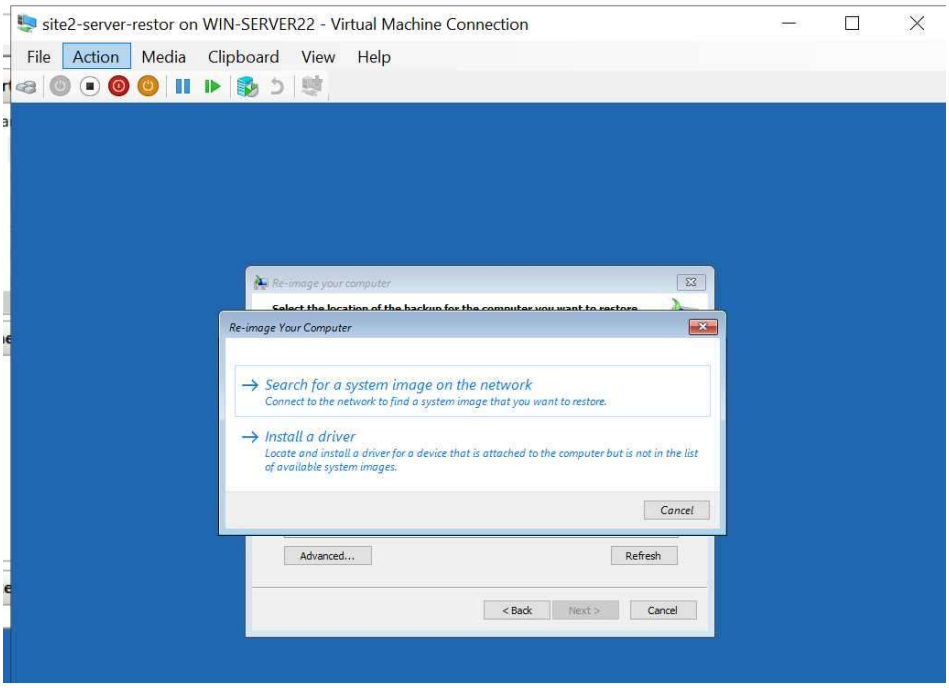


figure 24. screenshot promting you to choose where your system image is.

step6: Provide the path to the folder where the backup is stored (\\SITE2-BACKUP\Backups2) then click okay as shown in the figure 25 below..

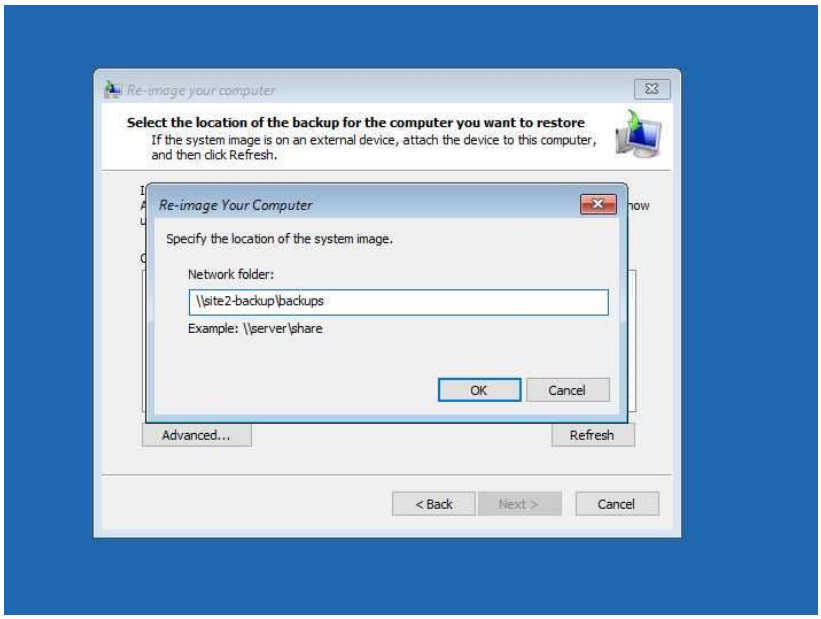


figure 25: snapshot of prompt to input the path to the backup file over the network.

*step7:*when prompted , provide credentials to log into the backup server (use an account that has permissions to access the server over the network). As shown in the figure 26 below. Scroll to okay and click. (note that you cannot use your mouse in this step)

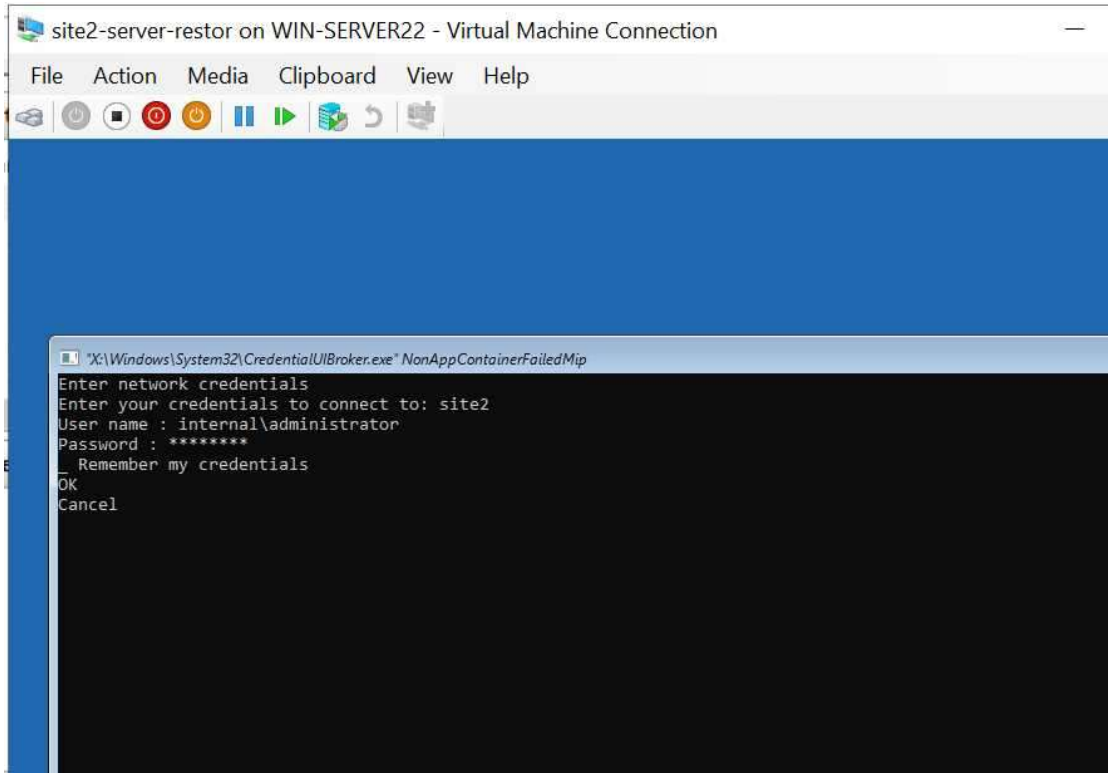


figure 26: wizzard asking for login credentials into the backup server

step8: select the backup image you want to restor and click okay.

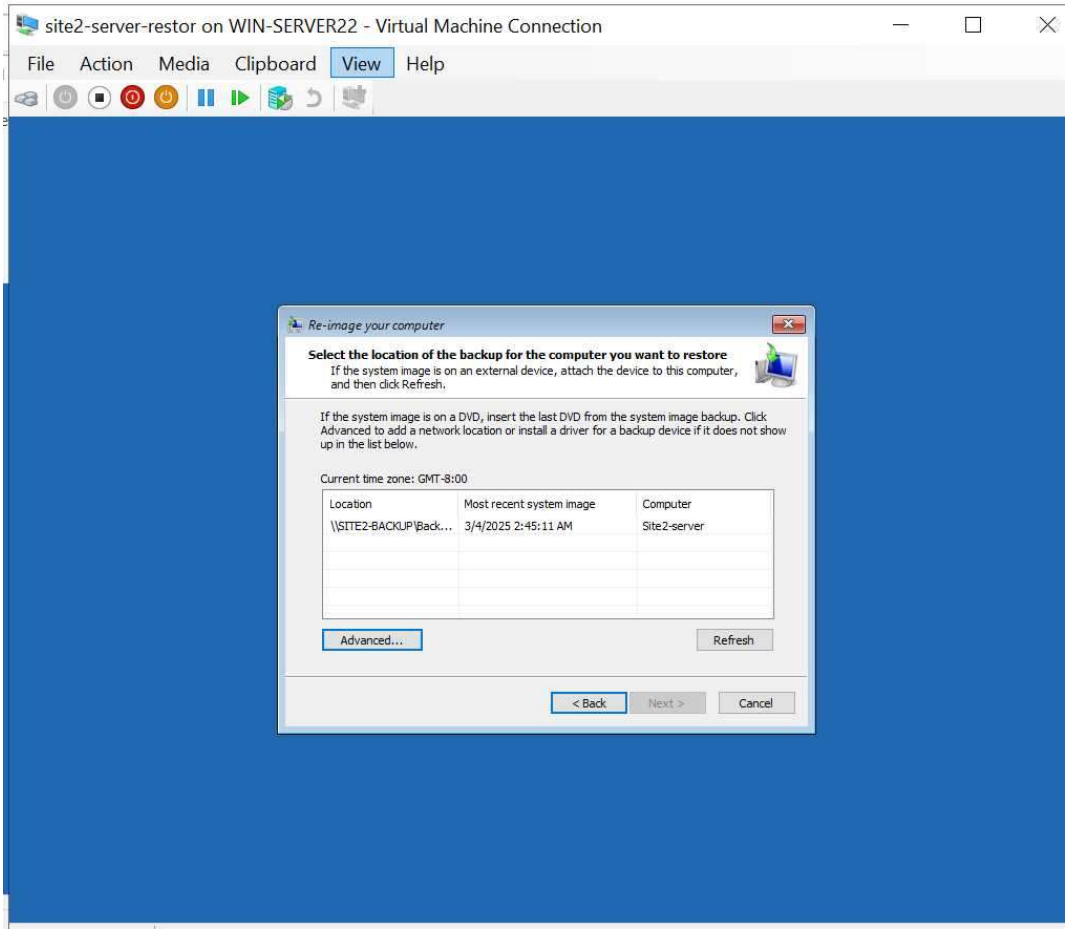


Figure 27: snapshot of wizzard showing the backup images found.

*Step9:*select the most recent backup image and click next as shown in fig 28.

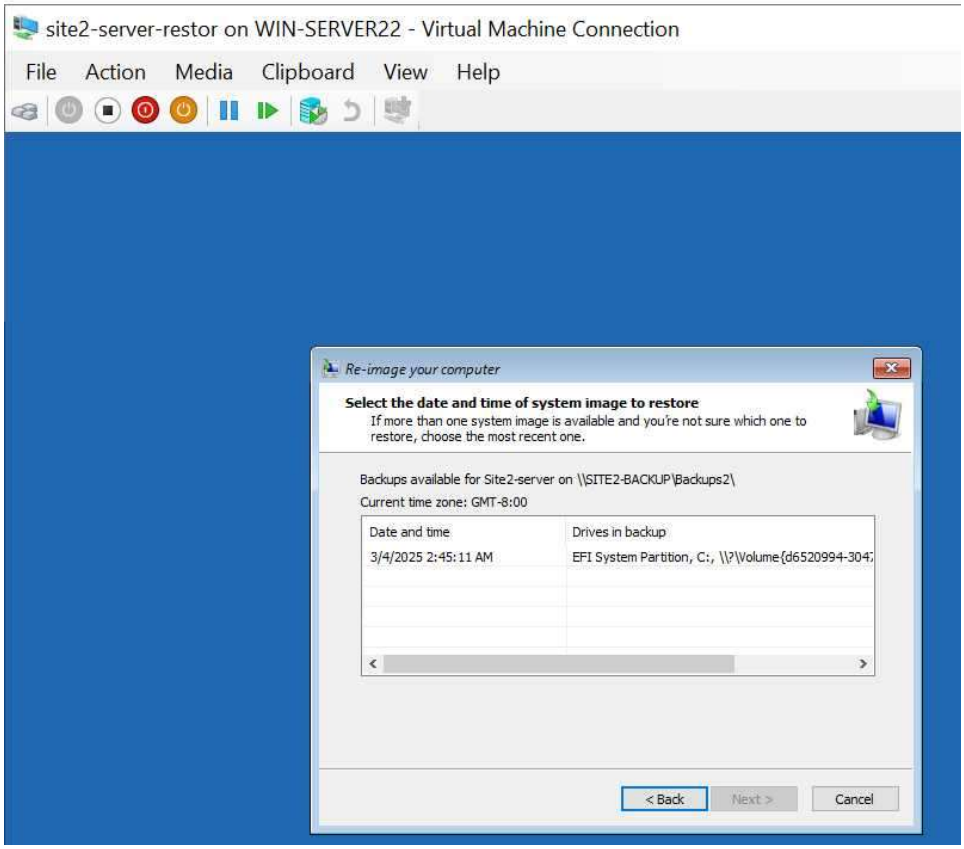


figure 28: Wizzard showing the date and time of all available backups.

step10: in figure 29, below you can choose perform additional restor configurations. In this case leave the default settings, click next.

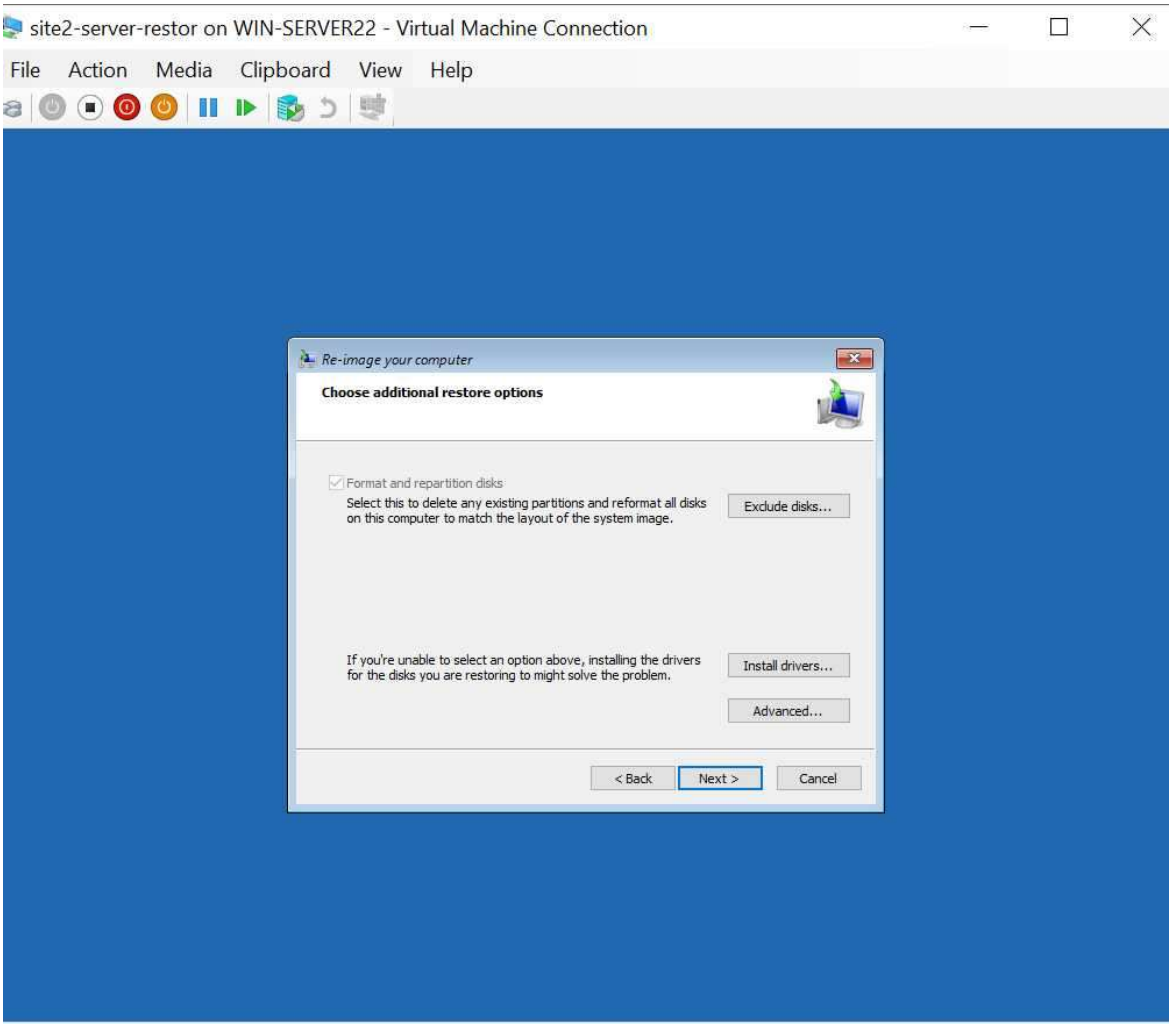


Figure 29: additional restor options wizzard.

*Step11.*click finish , then click yes on the notification that will pop up, to complete the restor process. See figure 30.

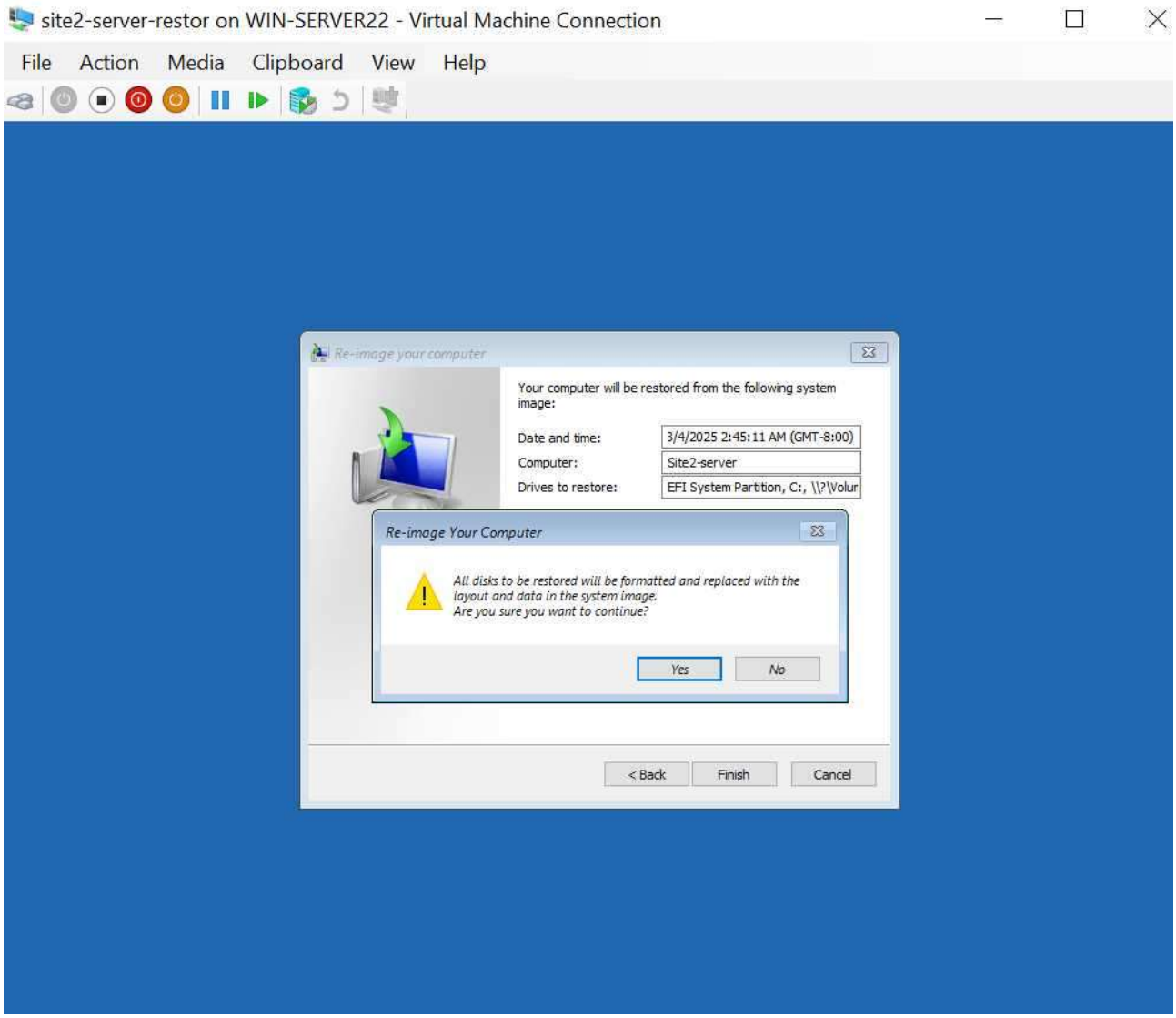


figure 30:snaption of the last step of the restoration process.

These steps will successfully perform a bare metal restoration of the site 2 servers. This server is an exact replica of the original server. However, the Site2-sever- restor has no internet connections.see figure 31 below.

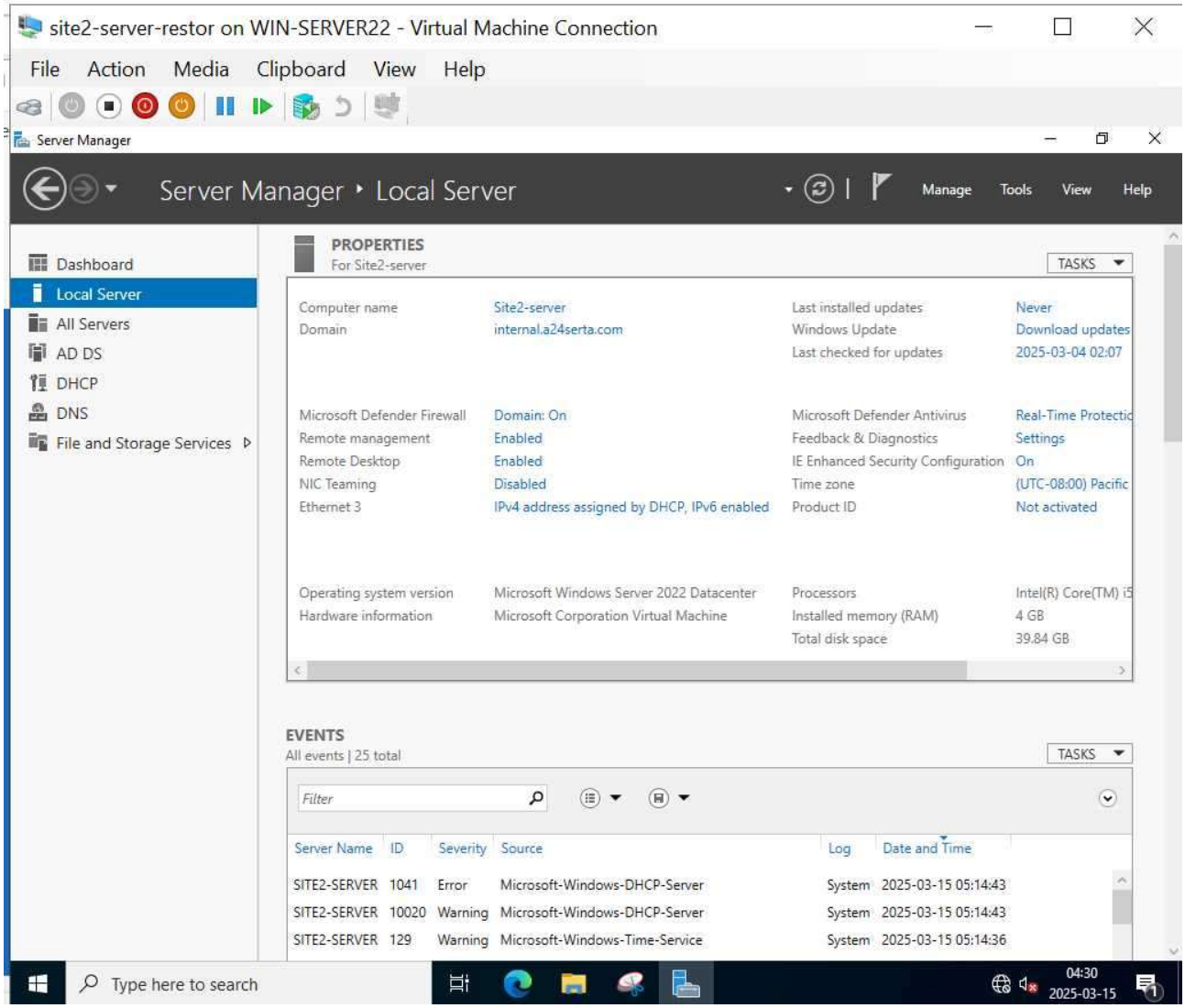


Figure 31: snapshot of the restored server running.

To connect this server to the internet, give it the same configure the ip adress to be the same as the original server. After this post configuration the server will automatically be connected to the internet.

2.Create a how-to guide that describes in detail how to create a new server from your master image by importing it into Hyper-V (pictures for each step is necessary).

Step1: Open the Hyper-v and click import virtual machine. Click netx on the import wizard as shown on the figure 32

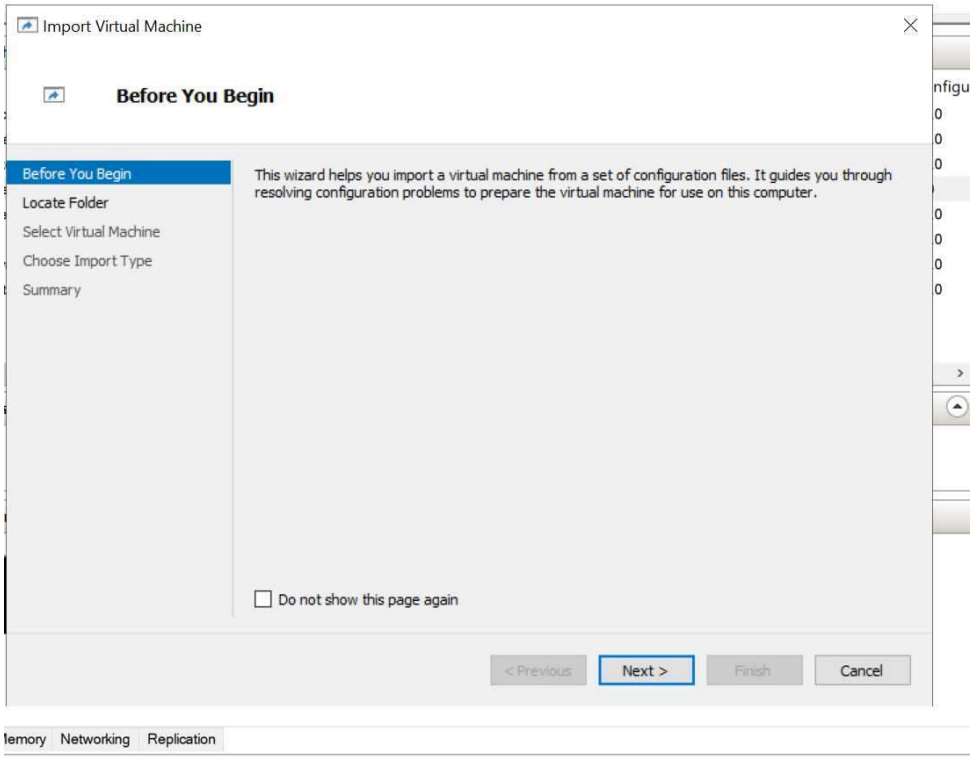


figure 32. import new virtual machine wizard.

step2: Specify the folder where the image server is saved in, click next

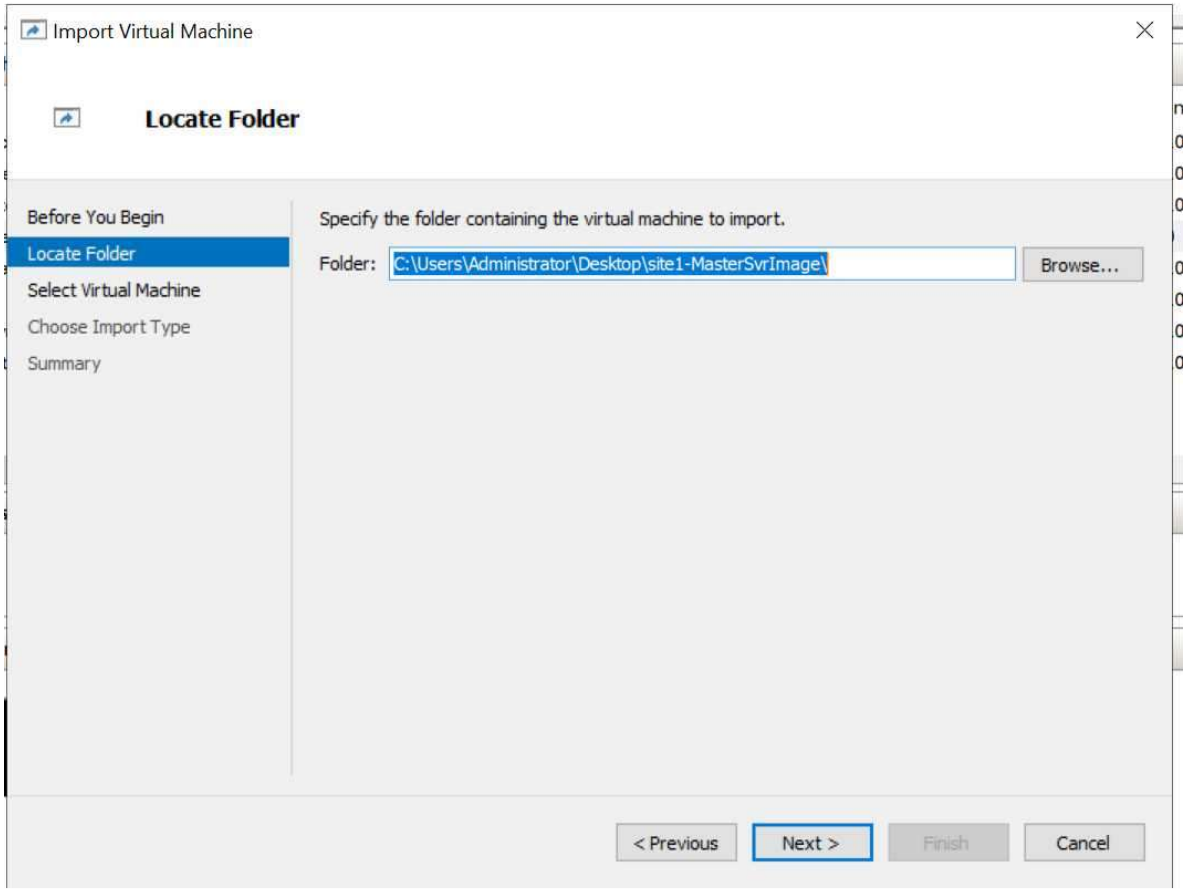


figure 33: Snapshot to locate the vm to be imported.

step3: select the image server you want to import and click next

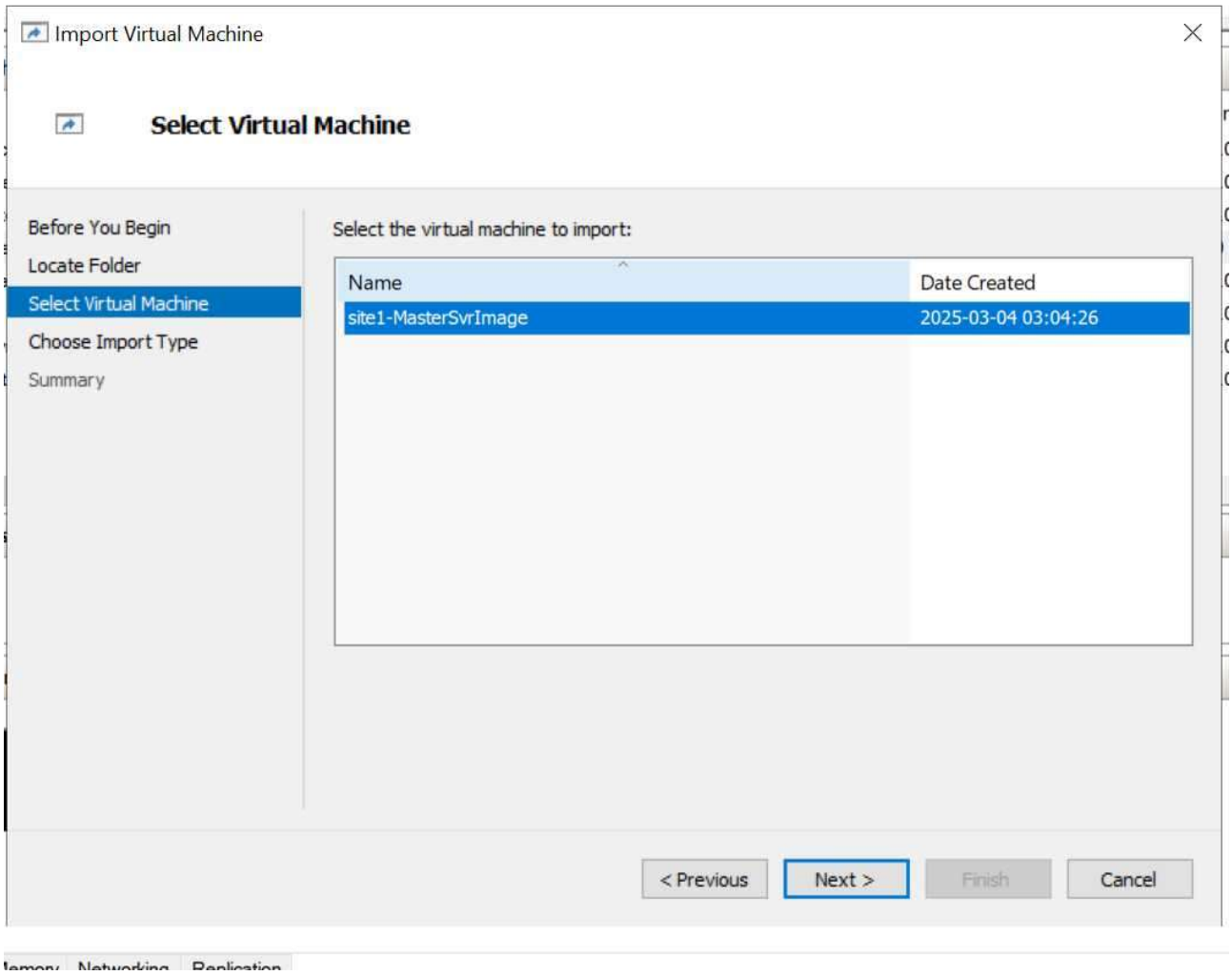


figure 34: snapshot of the vm to be imported

*step4: C*hoose the import type **copy the virtual machine** from the options as shown in figure 35, so that a new server is created with a unique iddentity. Click next.

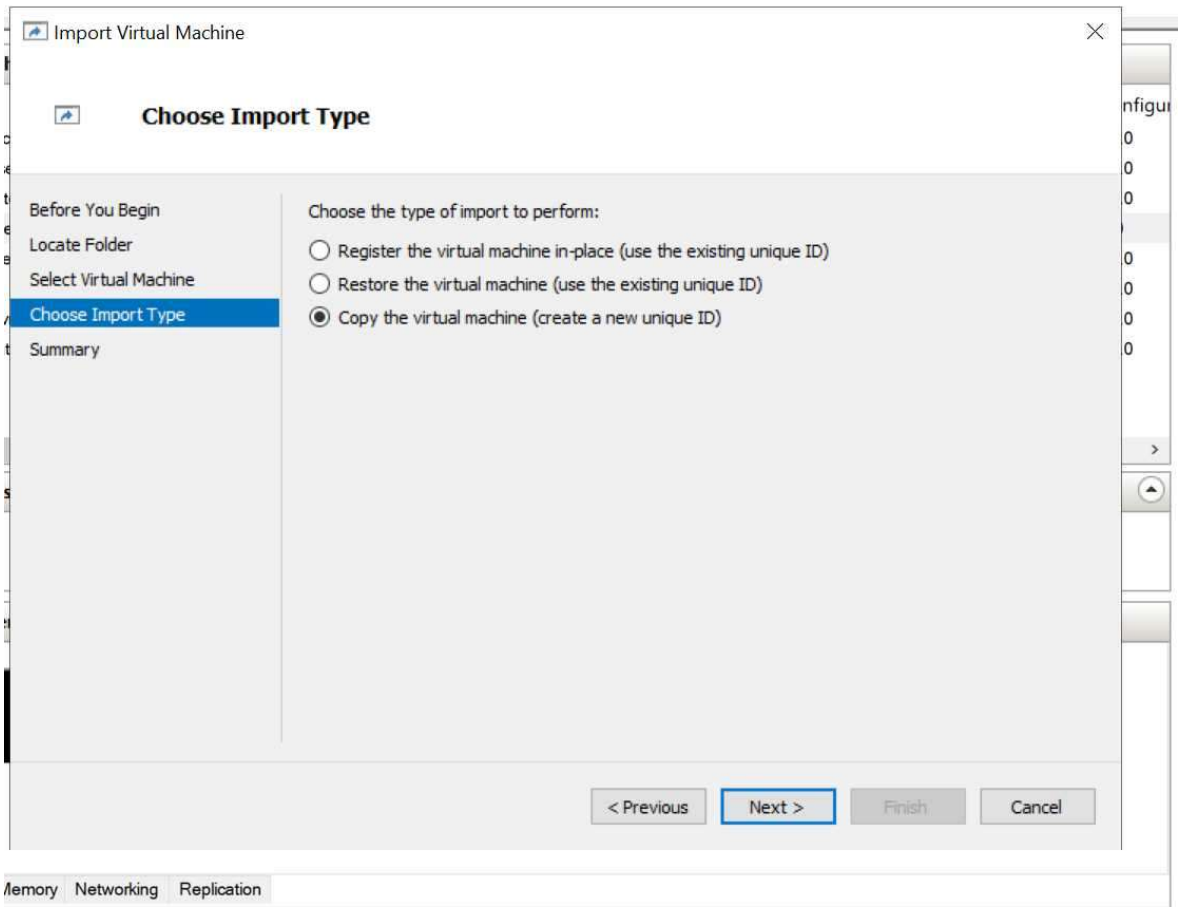


Figure 35:snapshot of import type options

Step5: choose folders to store the virtual machine files and click okay. As shown in figure , the files were saved in the default folders.

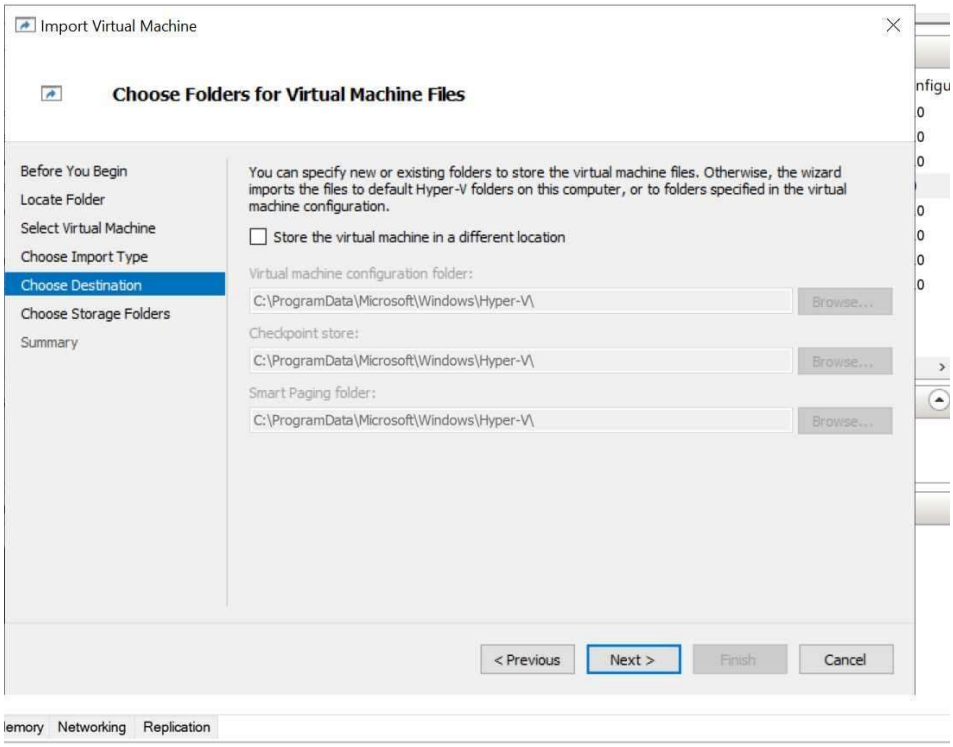


Figure 36:choose destination snapshot.

*Step6: C*hoose the folders to store the virtual hard disks. Note that 2 virtual hard disks of the same name cannot be stored in the same location *(that is why even after deleting a vm, you must brows to the folders where the hard disks was stored and delete it before you can successfully recreate another vm with the same name like the one deleted).* As shown in the fig 37, a new folder was created to store the imported vm's hard disk. Click next .

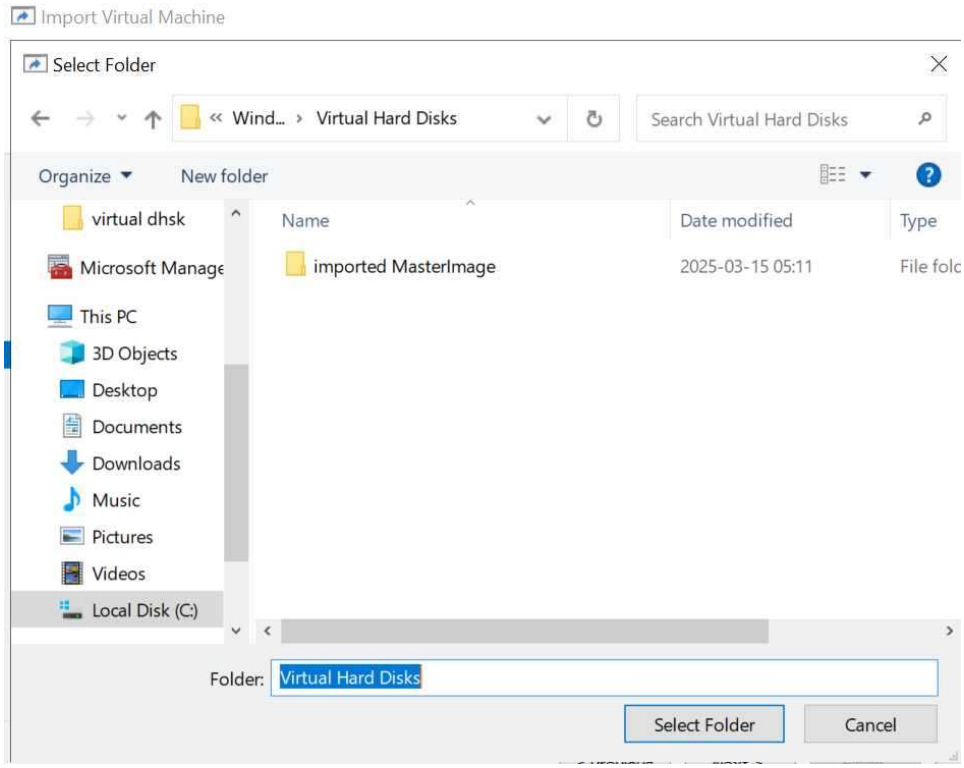


figure 37: Snapshot of the newly created folder to store the hard disk.

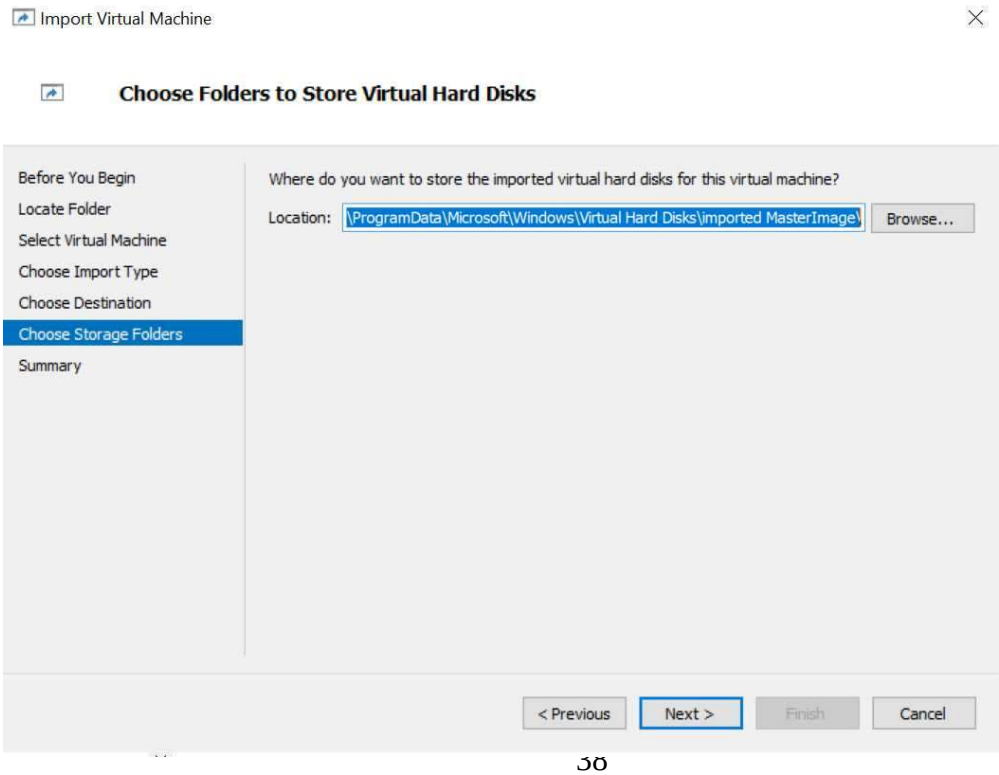


Figure 38: Snapshot showing the location where the hard disk has been stored

step7: click finish to complete the importation as shown in fig 39.

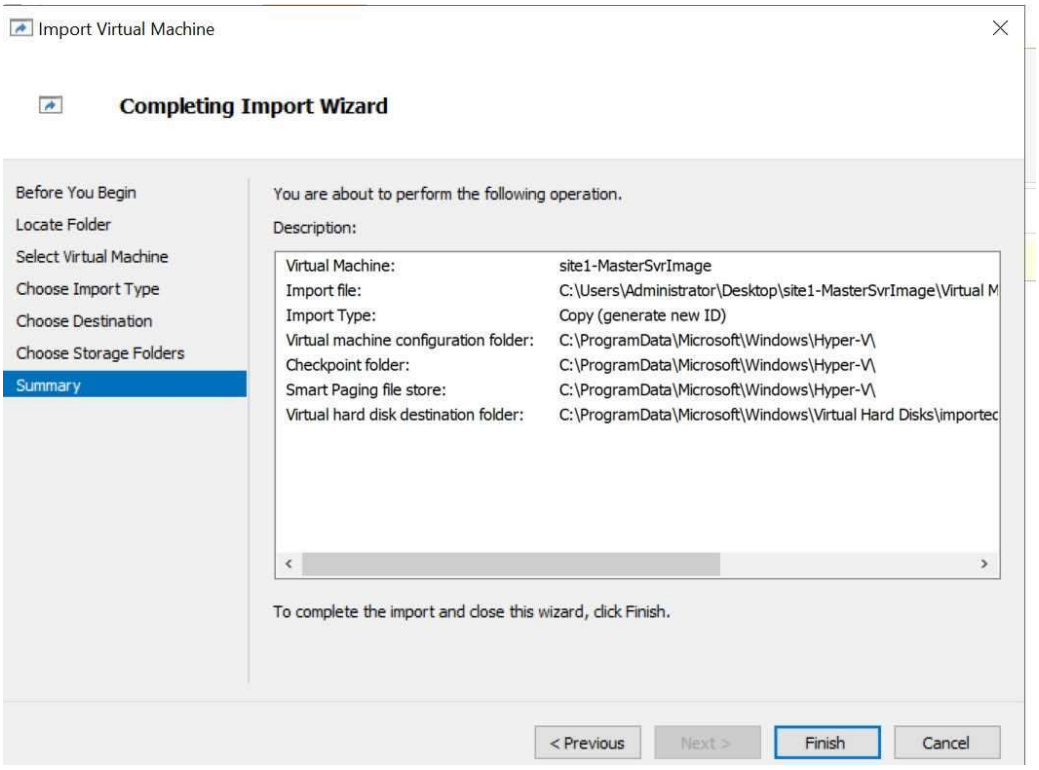


figure 39: Completing inportation wizard

Step8: Open settings for the imported vm and remove the DVD Drive as shown figure 40. Click apply then okay to apply changes .

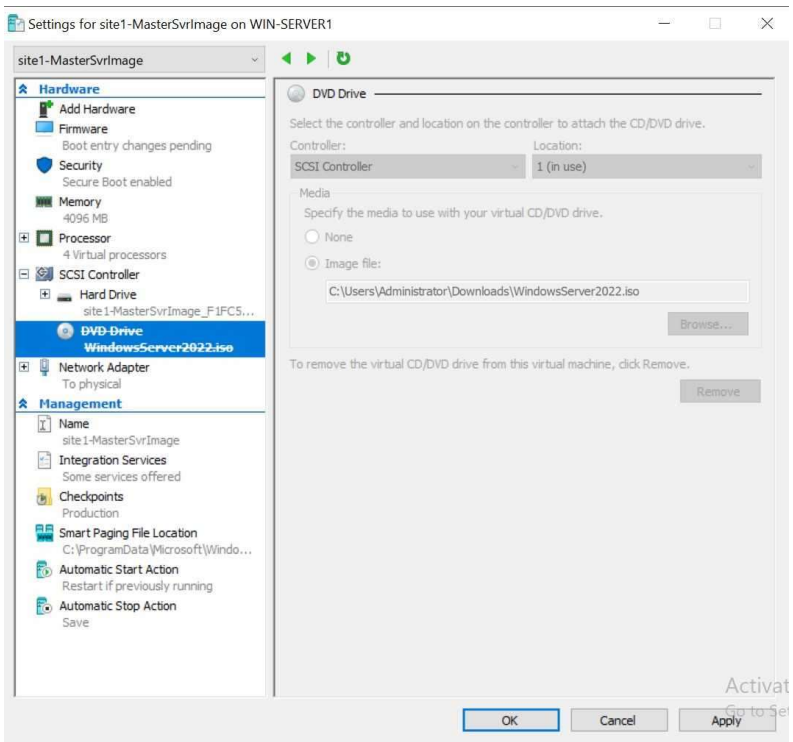


Figure 40:imported vm settings.

Step9: start the new vm, set your country, language and region (Sweden) and click next.

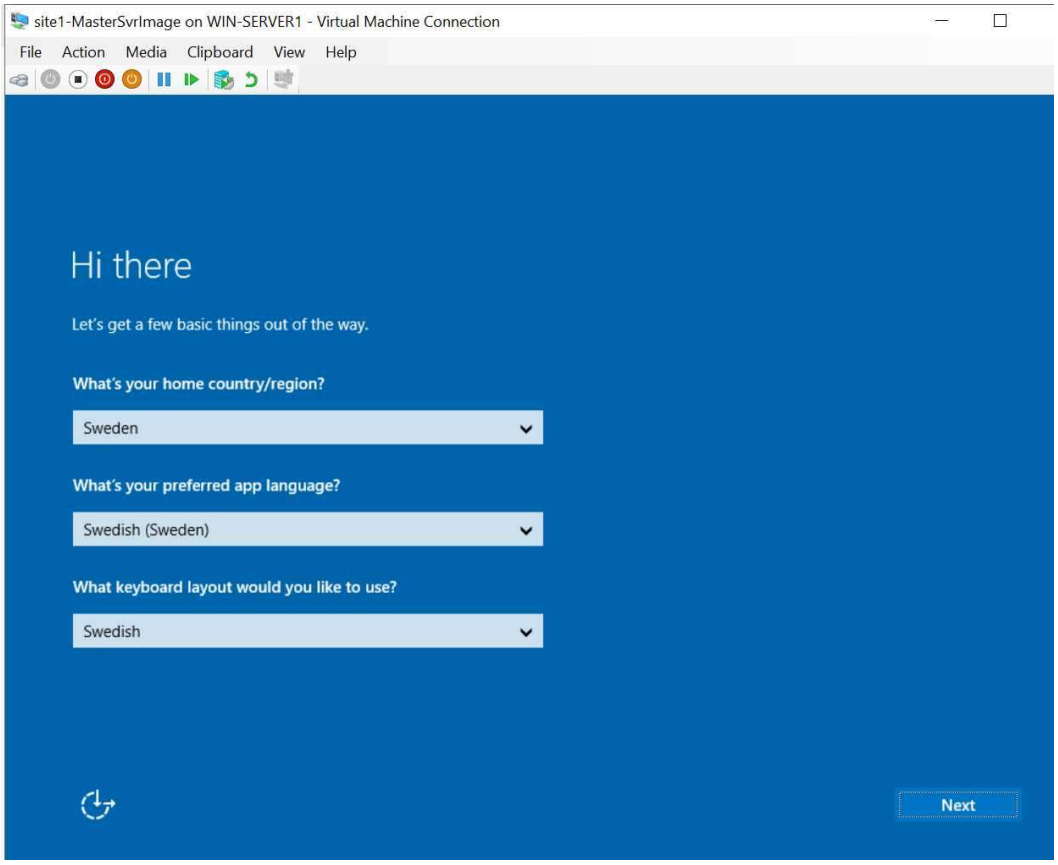


Figure 41: new vm start wizard snapshot

Step10: Wizard will then prompt you to input a product key. Since known was provided for this lab, click on do this latter (bottom left) then next and accept the license terms. See figure 42.

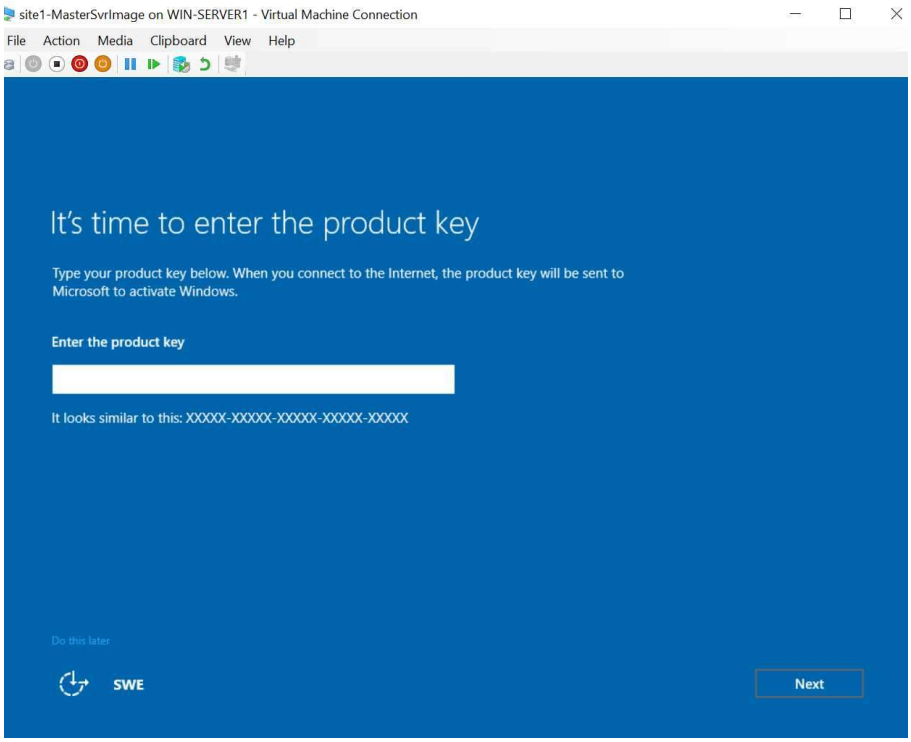


figure 42:product key prompt wizard snapshot

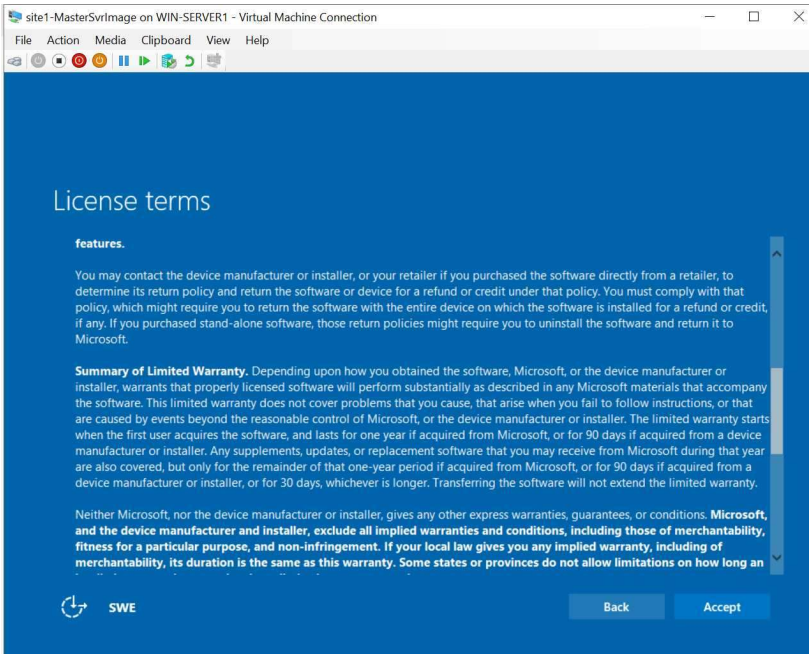


Figure 43: lisence terms wizard snapshot.

*step11: S*et the password for the administrator (Syp9393!) in the customize settings wizard, then click finish to complete the setup.

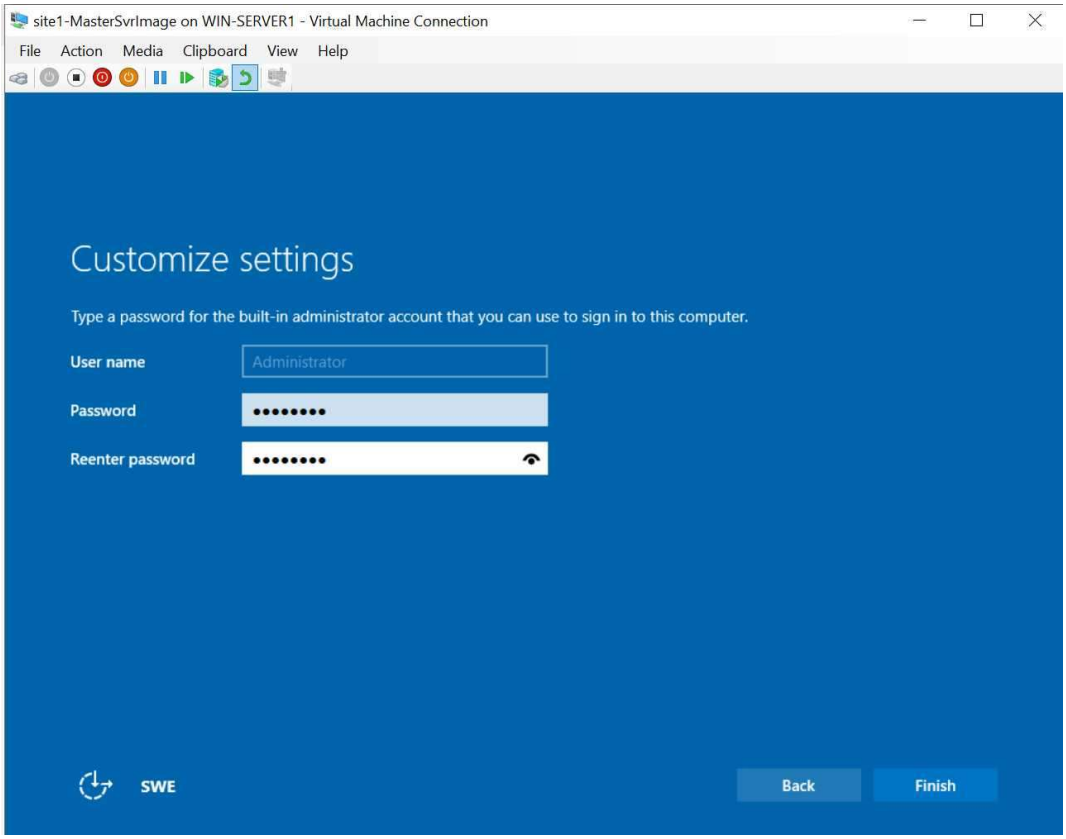


Figure 44: customize settings snapshot

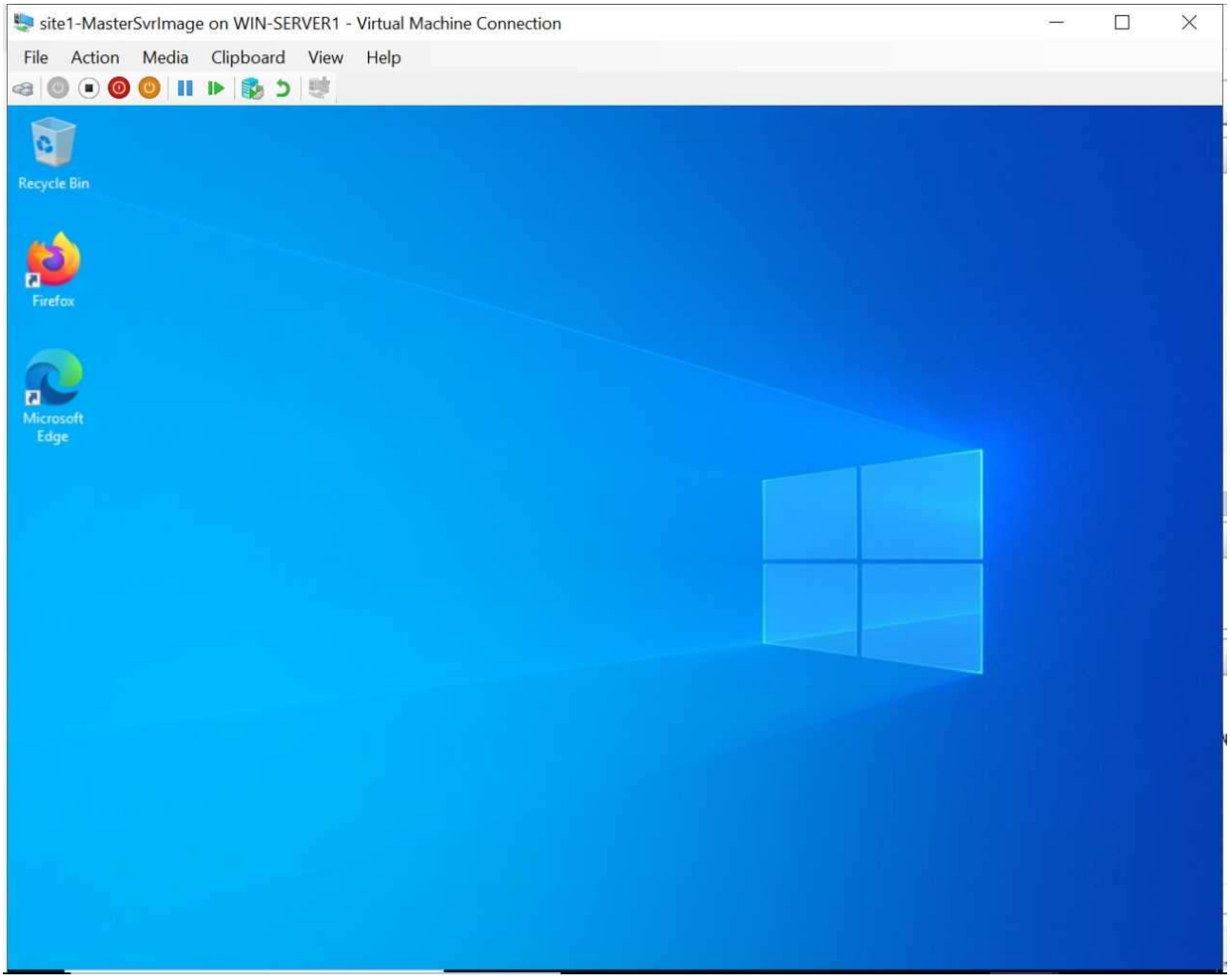


figure 45:newly created vm from the master server image.

Figure 45 above is a snapshot of the new machine created from the master image. This machine is connected to the internet and also has the firefox browser that was installed on the original browser.

3. Elaborate on the benefits of using a master image for deploying new servers

Section 8: Remote installation of software

1. Explain the difference between 'assign' and 'publish' that was used in this step?

We can install a software for a computer or users remotely from the GPO in Active Directory using either the assign or publish method. Applications can be assigned to both users and computer. When an application is assigned to a user or computer, it is automatically installed without any user intervention. For users, the application appears installed the next time they log in and computers, the installation happens at the next startup (Microsoft, 2025). This ensures that the software is readily available and enforced for the assigned users and computers.

On the other hand, publishing applies only to users. When an application is published, it becomes available for users to install via the Control Panel's "Programs and Features" section, but it is not installed automatically (Microsoft, 2025). This gives users the option to install the software themselves at any time, making publishing more flexible than assigning.

2. Where does the client find the software (VLC) that can be installed that we made possible using the MSI packages?.

As earlier discussed, when an application is published to the users, it is not installed automatically. The VLC was published to users and so for a user to be able to find this application.

- Login as an authenticated user to the client computer
- click the windows icon search for control panel. Open control panel and click programs
- click on install program from network.

3. Demonstrate how you immediately can uninstall the software from users and computers using GPO.

First you have to locate the software you want to uninstall. Remote installations can be configured for both users and computers . Inoder to locate the software follow the steps below.

- Select the policy you want to uninstall and click edit. Choose either computer configurations or user configurations based on the software.
- Expand policies Policies > Software Settings > Software Installation.
- In the right pane, select the software you want to uninstall, right click and choose all task.
- Under all task, click on remove. In the remove dialog box that will open, click on the option *Immediately uninstall the software from users and computers, then click okay to apply.*
- *Run gpupdate /force on the server to apply this change.*

Section 9: Automate deployment of new client computers

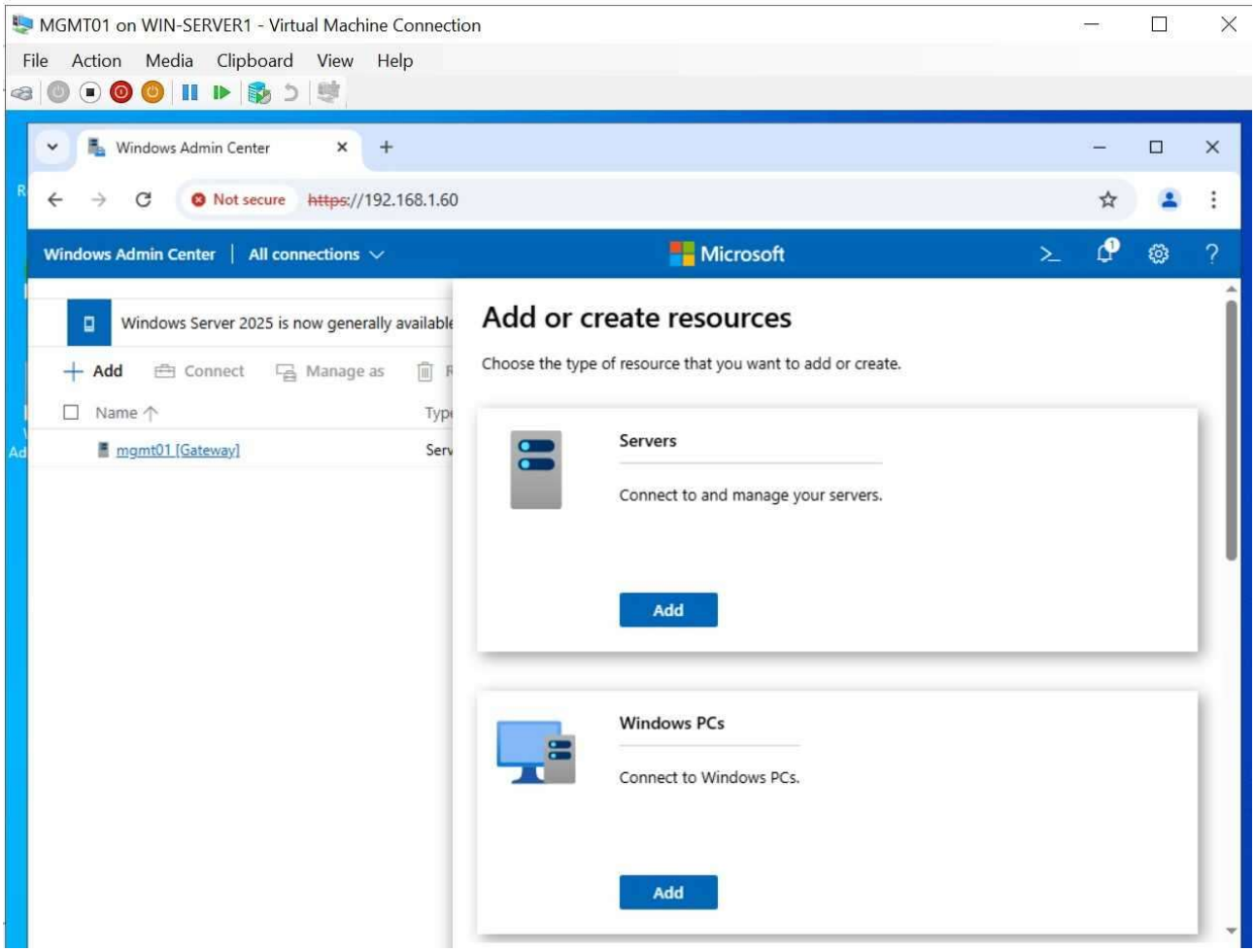
1. What protocol is used to transfer the operating system image during the PXE deployment?.

In the Preboot Execution Environment (PXE), the Trivial File Transfer Protocol (TFTP) is primarily used to transfer the initial boot files, such as the Network Bootstrap Program (NBP), from the server to the client (Microsoft n.d; wikipedia, 2024). After the NBP is loaded, the client may utilize more robust protocols like HTTP, HTTPS, SMB, or NFS to download larger operating system images, enhancing the efficiency and reliability of the transfer process (Wikipedia, 2024)

Section 10: install management tools

1. How you add servers to WAC?

Step1: log into the WAC, click add and choose servers as the resources yo want to add.


MGMT01 on WIN-SERVER1 - Virtual Machine Connection
File Action Media Clipboard View Help
Windows Admin Center
Not secure https://192.168.1.60
Windows Admin Center | All connections
Microsoft
Windows Server 2025 is now generally available
Add Connect Manage as
Name
mgmt01 [Gateway]
Add or create resources
Choose the type of resource that you want to add or create.
Servers
Connect to and manage your servers.
Add
Windows PCs
Connect to Windows PCs.
Add
Figure 46

Step 2: To add many servers, click search active directory and type ”*”.and click search. This will find all the servers in the active directory. Ckeck the box infront of name to sellect all the servers then click add. All the server will be added to your list under all connections. Click on any server to coonnect.

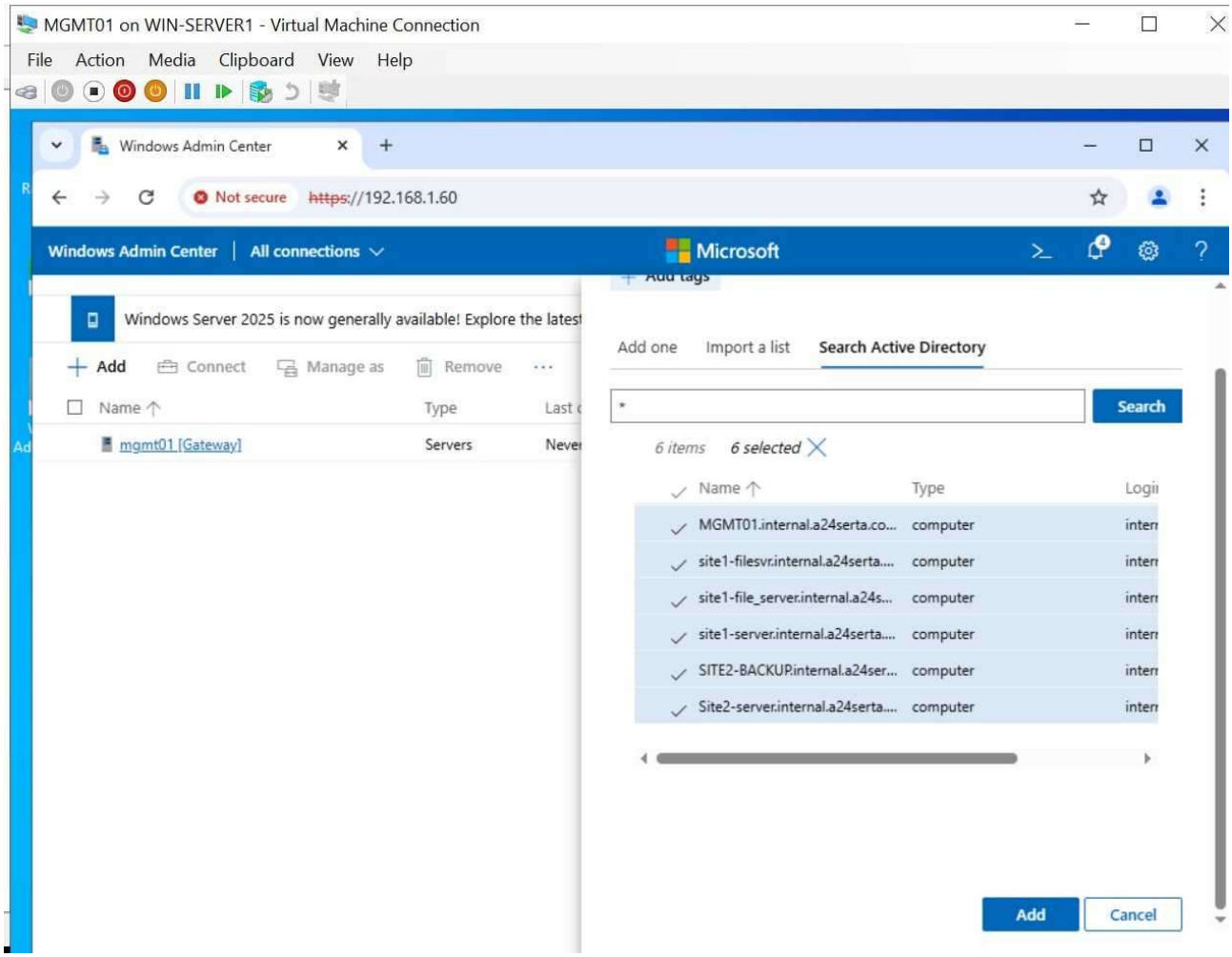


Figure 47

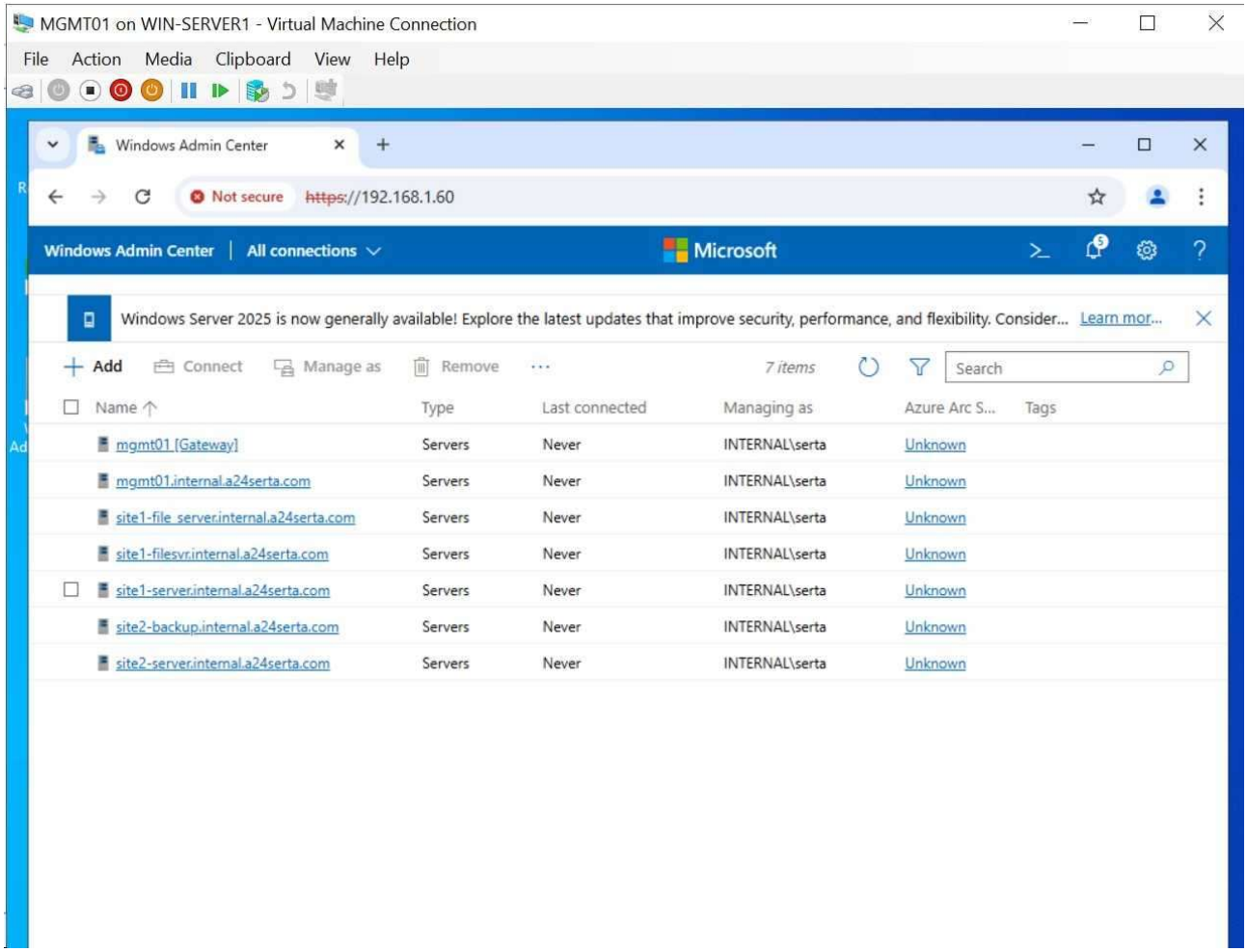

MGMT01 on WIN-SERVER1 - Virtual Machine Connection
File Action Media Clipboard View Help
Windows Admin Center
Not secure https://192.168.1.60
Windows Admin Center | All connections
Microsoft
Windows Server 2025 is now generally available! Explore the latest updates that improve security, performance, and flexibility. Consider... Learn mor...
Add Connect Manage as Remove
7 items
Search
Name Type Last connected Managing as Azure Arc S... Tags
mgmt01 [Gateway] Servers Never INTERNAL\serta Unknown
mgmt01.internal.a24serta.com Servers Never INTERNAL\serta Unknown
site1-file_server.internal.a24serta.com Servers Never INTERNAL\serta Unknown
site1-filesvr.internal.a24serta.com Servers Never INTERNAL\serta Unknown
site1-server.internal.a24serta.com Servers Never INTERNAL\serta Unknown
site2-backup.internal.a24serta.com Servers Never INTERNAL\serta Unknown
site2-server.internal.a24serta.com Servers Never INTERNAL\serta Unknown

figure 48

2. How to connect to a server via RDP using WAC.

Step1.select the server and click connect

step2: after connecting to the server, search for remote desktop under tools and click on the icon ticked on the figure below to open on a new windows.

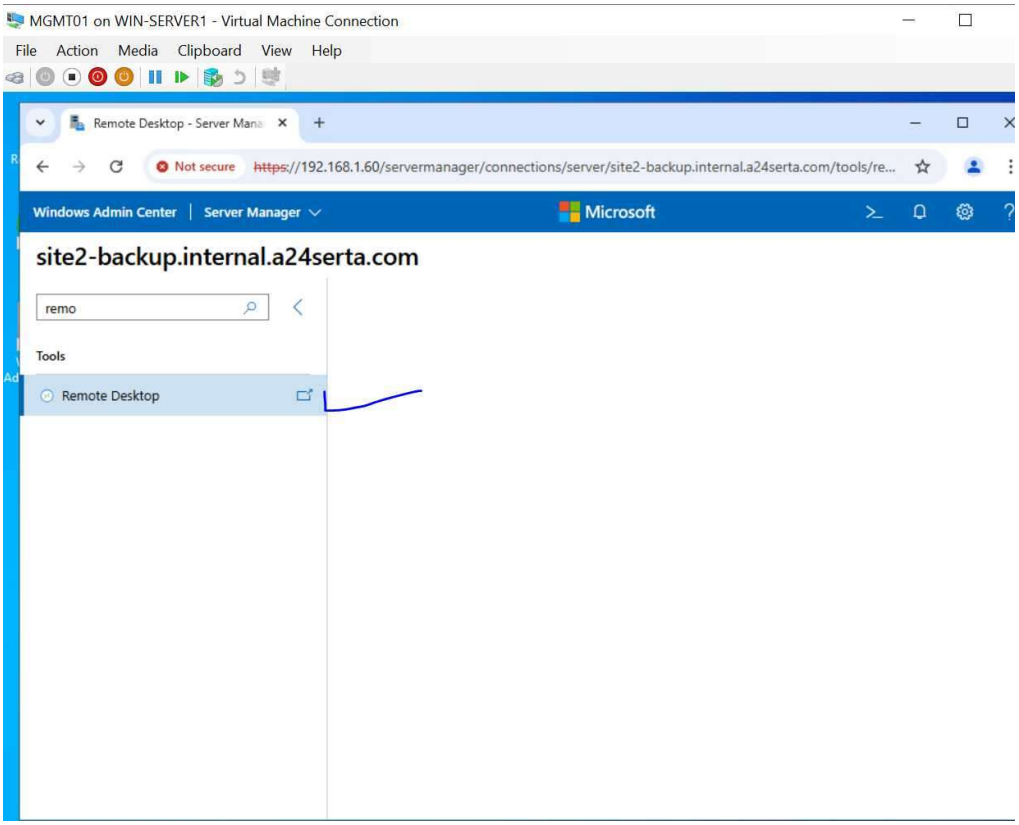


Figure 49

Step3: On the remote desktop window that will open, provide a valid password, choose the keyboard and check *automatically connect with the certificate presented by this machine* as shown on figure 50. click confirm on the next wizard as shown in figure 51.

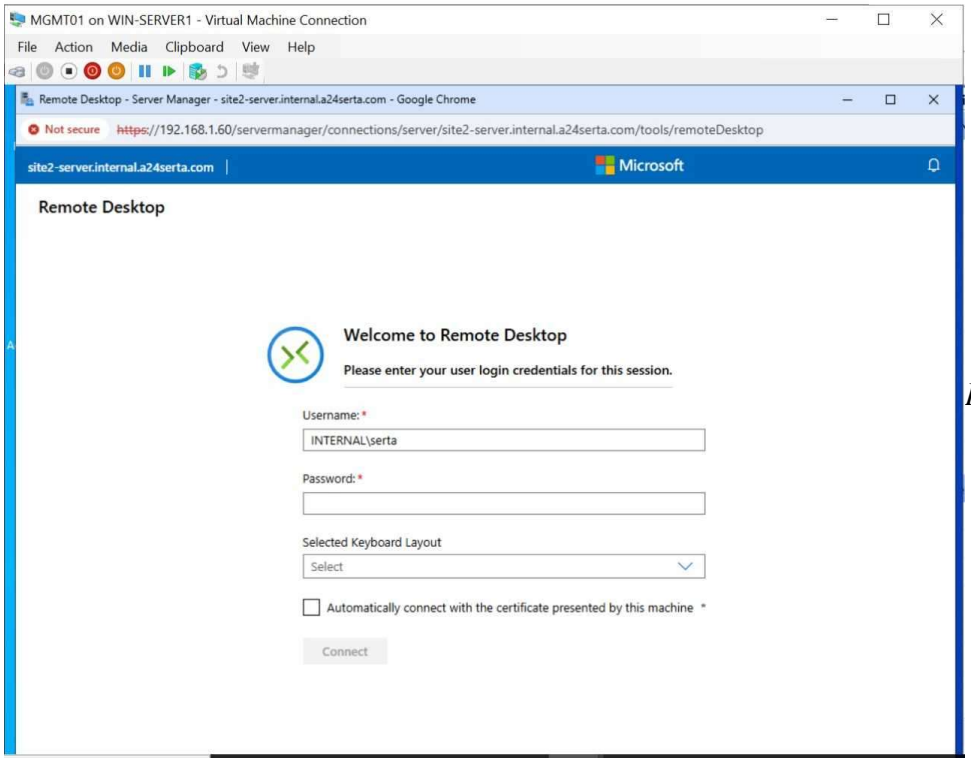


Figure 50.

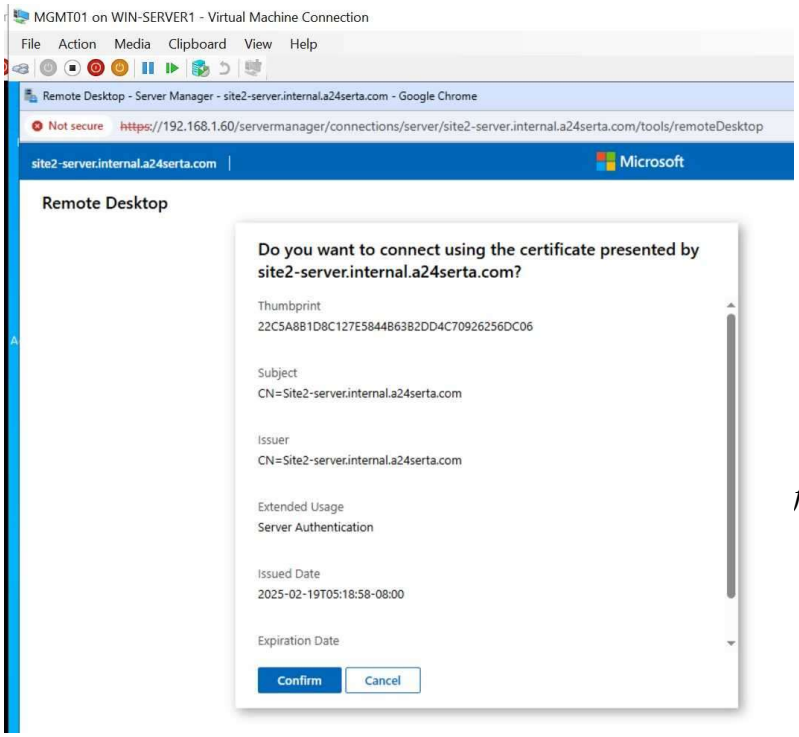


figure 51:snaption to confirm certificate.

Step4: On the windows logon screen that will appear, provide the login credentials of a user with RDP access permission. For this lab, members of the IT departement were granted RDP access to site1-server and site2-server. As seen in figure 52, user joka was used to log in to the server via RDP.

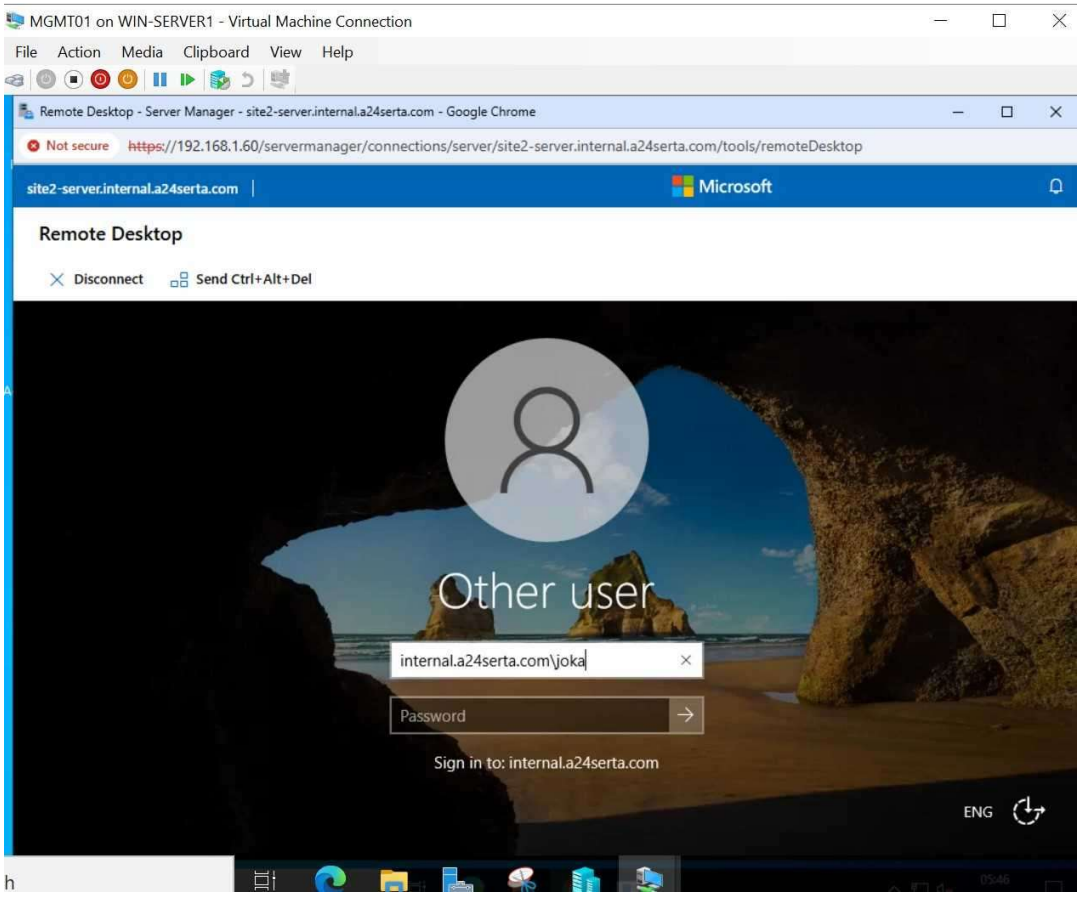


Figure 52: login screen.

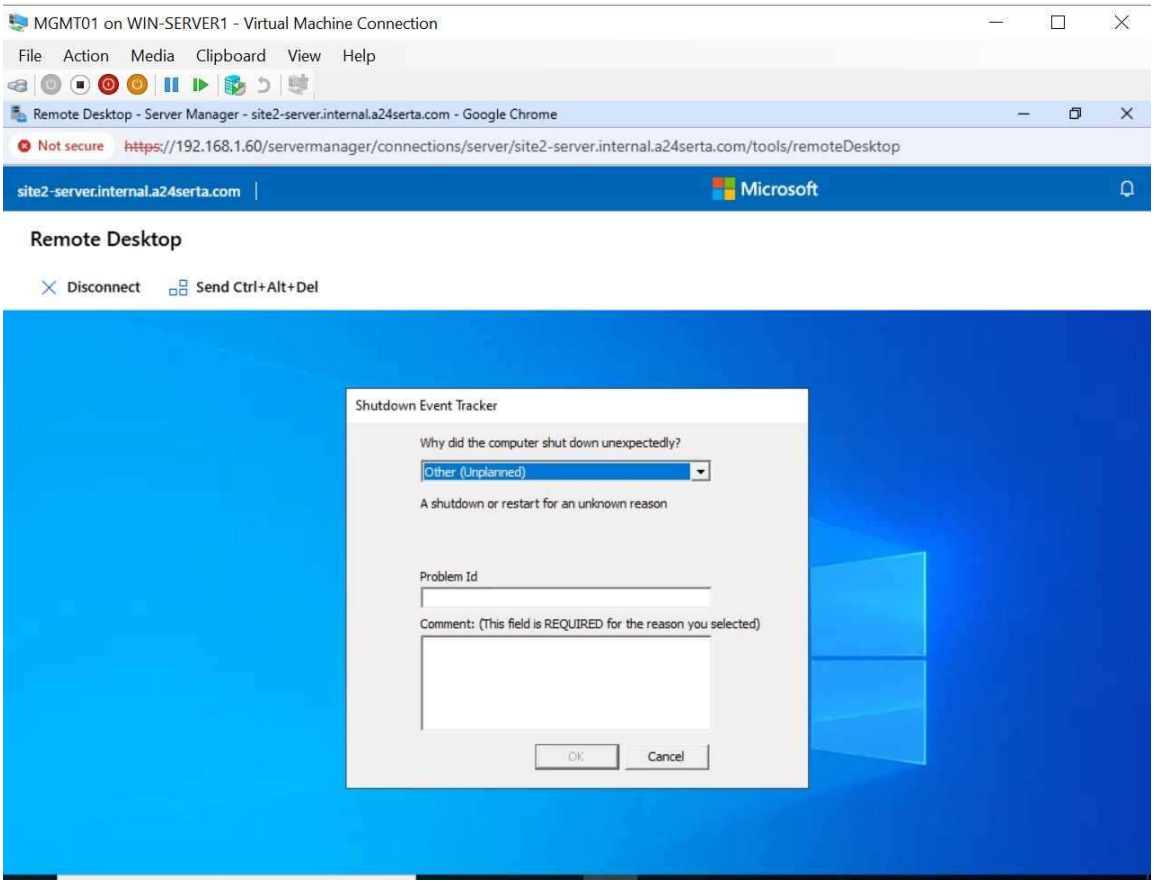


Figure 53:site1-server, login via RDP on WAC.

3. The link to the official documentation on WAC?

https://www.microsoft.com/en-us/evalcenter/download-windows-admin-

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